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Inference for partially observed competing risks model for Kumaraswamy distribution under generalized progressive hybrid censoring

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ABSTRACT

In this paper, inference for a competing risks model is studied when latent failure times follow Kumaraswamy distribution and causes of failure are partially observed. Under generalized progressive hybrid censoring, existence and uniqueness of maximum likelihood estimators of model parameters are established. The confidence intervals are obtained by using asymptotic distribution theory. We further compute Bayes estimators along with credible intervals. In addition, inference is also discussed when there is order restricted shape parameters. The performance of all estimates is investigated using Monte-Carlo simulations. Finally, analysis of a real data set is presented for illustration purposes.

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1. Introduction

Often competing risks models play significant role in reliability and life testing as several causes may affect failures of different test units. Such studies are also useful in mortality data analysis. In fact, multiple causes may lead to failures depending upon internal and external structure of units under study, their design pattern, manufacturing process etc. Under competing risks context observations are commonly derived along with corresponding cause of failures known as competing risks. Thus data usually may consist of failure times and an associated indicator variable indicating causes of failure. Competing risks models have received extensive attentions in literature, see for instance, Kundu and Pradhan [15], Cramer and Schmiedt [7], Wang [39,41], Koley and Kundu [13], for some recent contributions. Among others, Crowder [8] and Pintilie [31] have also discussed some interesting applications of competing risks models. Most of the above references analyzed such models when causes of failures are observed for each of the failed experimental unit. However, it may happen that failure causes are not observed for each unit but only for a portion of

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failed experimental units in practical situations. For example, suppose a group of male mice are subjected to some radiation dose in a laboratory and cause of deaths are classified as Thymic Lymphoa and Reticulum Cell Sarcoma. But in real situation, the death may happen due to some unknown unidentified causes. Such competing risks can be classified as partially observed competing risks. Therefore, it is of practical significance and interest to discuss the inference problems in many investigations when there are partially observed causes of failure. In many fields of practical studies, it is often observed that one failure cause may be dominating the other mode for the failure of some systems or components. Such situations may affect the expected life of experimental subjects owing to different modes of failure. Inferences related to identification of failure causes are quite useful for improving product quality and safety related issues as well. Systematic measures against reliability may lead to develop guidelines for system maintenance procedures also. We further mention that order restricted inferences have found wide applications in various life test studies. This type of inference problems are also useful in the study of agricultural, industrial and bio-assay experiments. Consider an accelerated life testing situation where experimental units are tested under different stress levels. It is known that with an increase in the stress level more units will fail. Thus units under higher stress levels will have smaller lifetimes compared to the corresponding lifetimes under relatively smaller stress level. In a mortality analysis different groups of animals are given varying doses of a treatment. More animals may die with an increase in the drug dosage. So it is expected that proportions of animals died using large dose will be more compared to the lesser dosage. Thus it is natural to incorporate such additional information in an inference problem to derive better procedures for unknown parameters.

Because of constraints associated with a life testing facilities, lifetimes of all units put on a test are often not observed and so data are censored in nature. Censoring arises quite frequently in life testing studies. The two widely used censoring are Type-I and Type-II censoring schemes. One of the important features of these schemes is restriction on removing live units during the test. Then the progressive censoring scheme (PCS) is further introduced in practice which is more flexible and allows engineers to remove live units at different test stages. We refer to Kundu [17], Sen *et al.* [35], Pradhan and Kundu [32], Dey and Dey [9], Singh *et al.* [37], Maurya *et al.* [22], Lodhi *et al.* [20], Mahto *et al.* [21] for some recent applications. One may also refer to Balakrishnan and Aggarwala [1] and Balakrishnan and Cramer [2] for an excellent discussion on this censoring methodology. In literature, Type-I and Type-II hybrid censoring are also introduced to overcome drawbacks like no failure under prefixed testing period, cost under long testing periods etc. Type-I hybrid censoring is a mixture of Type-I and Type-II censoring where a life test stops at a prefixed time or when a prefixed number of failures is observed, whichever occurs first. In Type-II hybrid censoring, a test is terminated at the later occurrence of the previous two events. Some of useful references on hybrid censoring are Childs *et al.* [5], Kundu [17], Rastogi and Tripathi [33] and Sultana *et al.* [38]. The above discussed censoring schemes may not simultaneously control testing period and number of failures in an experiment. To overcome such situations, progressive hybrid censoring has been introduced which is a mixture of hybrid censoring with additional characteristic that live units can be withdrawn at various stages of an experiment. We refer to Kundu and Joarder [14], Childs *et al.* [4], Lin *et al.* [18,19], Sen *et al.* [34] and references cite there-in for applications of this censoring. Recently, Cho *et al.* [6] introduced and studied a generalized progressive hybrid censoring

(GPHC). Suppose n identical units are placed on a life testing experiment. Also, let k and m are fixed with $1 \leq k < m \leq n$ and value of T is fixed as well. The censoring scheme $R_i \geq 0,\ i = 1, \ldots, m$ satisfies $\sum_{i=1}^{m} R_i = n - m$. As in PCS, at ith failure time, R_i number of units are removed and the failure $Y_{i:m:n}$ is recorded. A test under GPHC is terminated at $T^* = \max\{Y_{k:m:n}, \min\{T, Y_{m:m:n}\}\}$. Notice that GPHC provides flexibility to continue an experiment beyond the time point T if adequate number of failures are not observed.

Let $Y_{ji},\ i = 1, \ldots, n$ denote the failure time of ith unit which is failed due to the jth cause. Suppose latent failure times follow Kumaraswamy distributions with shape parameters $\lambda_1\lambda_2 > 0,\ \lambda_1 \neq \lambda_2$ and a common shape parameter α. The probability density function (PDF) and the cumulative density function (CDF) of this model are given as

$$f_j(y) = \alpha\lambda_j y^{\alpha-1}(1 - y^{\alpha})^{\lambda_j - 1}, \quad \alpha, \lambda_j > 0,\ 0 < y < 1; \tag{1}$$

and

$$F_j(y) = 1 - (1 - y^{\alpha})^{\lambda_j}, \quad \alpha, \lambda_j > 0,\ 0 < y < 1; \tag{2}$$

respectively. Kumaraswamy [16] initially studied this distribution for modeling hydrological data such as daily rainfall and daily stream flow. Its density function can be unimodal, uniantimodal, increasing, decreasing or constant depending upon its range of parameters. For instance, for $\alpha > 1,\ \lambda_j > 1$ density function is decreasing, $\alpha < 1,\ \lambda_j < 1$ corresponds to uniantimodal case, $\alpha > 1,\ \lambda_j \leq 1$ lead to increasing density, $\alpha \leq 1,\ \lambda_j \leq 1$ and $\alpha = \lambda_j = 1$ corresponds to decreasing and uniform densities, respectively. The quantile function has nice explicit expression as well. One may further refer to Jones [12] for a nice comparison of beta distribution with the Kumaraswamy distribution. Various properties and the advantages of Kumaraswamy distribution over beta distribution are explained in detail. Also, it is well explained that any phenomena with some upper and lower bounds can be modeled using this distribution, since this distribution can be transformed to any other boundary support (L, U) by simple linear transformation $L + (L - U)X$ of a random variable X. Gholizadeh *et al.* [11] have obtained the point and interval estimates under classical as well as Bayesian set up with progressively type-II censored data for this distribution. Further, Sindhu *et al.* [36] has considered the classical and Bayesian estimation and prediction under failure censoring sampling for this distribution. The study of Kumaraswamy distribution is also extended to stress–strength model. In this regard, one may refer to Nadar *et al.* [28] who have considered estimation of stress–strength reliability from both classical as well as Bayesian contexts. The study of stress–strength reliability is also considered under record values by Nadar and Kizilaslan [27]. For some more discussion of applicability of this distribution, one may also refer to Nadarajah [26], Sultana *et al.* [38] and Wang [40].

From literature we note that phenomena with lower and upper bounds can be modeled using Kumaraswamy distribution. Therefore, many of the lifetime phenomena such as human mortality, electrical and electronic devices and medical treatments may have some lower and upper lifetime bounds and thus this distribution can be used as failure time model. Motivated by the applicability of this distribution from previous studies and a more realistic scenario of partial observation of failure causes in study of competing risks, our main aim is to derive various statistical inferences under GPHC scheme when failure lifetimes of experimental units follow Kumaraswamy distribution with partially observed failure causes. Under arbitrary and order restricted shape parameters λ_1 and λ_2, different point and interval estimators of unknown parameters are discussed when partially

observed competing risks data are available. We further mention that extension of complete competing risks situation to a practical scenario of partially observed failure causes when latent failure times follow Kumaraswamy distribution is of significant interest in life testing inferences. This probability model is widely used in many physical phenomena where observations lie in a finite range like, average per capita income, average fuel efficiency, heights of individuals, scores on aptitude test, hydrological data etc.

The rest part of this paper is organized as follows. Model description and prior assumptions are presented in Section 2. Estimation without order restriction is considered in Section 3. In Section 4, classical and Bayesian estimation are considered with order restriction on shape parameters λ_1 and λ_2. Equivalence testing is discussed in Section 5. Monte Carlo simulations are performed to examine the behavior of all estimators and analysis of a real data set is discussed in Section 6. Finally, a brief concluding remark is presented in Section 7.

2. Model description and priors

2.1. Model description

Suppose a life testing experiment starts with $n \in \mathbb{N}$ identical units and failure times of these units are given by $Y_1, Y_2, \ldots, Y_n$. With two cause of failures, the latent failure times of the ith unit due to the jth cause of failure is denoted by Y_{ji}, $j = 1, 2$ where Y_{1i} and Y_{2i} are independent, then $Y_i = \min\{Y_{1i}, Y_{2i}\}$, $i = 1, 2, \ldots, n$. Quantities T and k denoting time and number of failures, respectively, are also fixed in advance along with censoring scheme $R_1, R_2, \ldots, R_m$. The test stops at following time

$$T^* = \max\{Y_{k:m:n}, \min\{T, Y_{m:m:n}\}\} \tag{3}$$

re-expressed as

$$T^* = \begin{cases} Y_{k:m:n}, & \text{if } T < Y_{k:m:n} < Y_{m:m:n}, \\ T, & \text{if } Y_{k:m:n} < T < Y_{m:m:n}, \\ Y_{m:m:n}, & \text{if } Y_{k:m:n} < Y_{m:m:n} < T, \end{cases} \tag{4}$$

where $Y_{1:m:n}, Y_{2:m:n}, \ldots, Y_{m:m:n}$ are progressively censored data corresponding to Y_1, Y_2, $\ldots$, Y_m. The competing risk data under GPHC is now given as:

$$\begin{cases} \text{CaseI}: & (Y_{1:m:n}, \delta_1, R_1), (Y_{2:m:n}, \delta_2, R_2), \ldots, (Y_{k:m:n}, \delta_k, R_k), & \text{if } T < Y_{k:m:n} < Y_{m:m:n}, \\ \text{CaseII}: & (Y_{1:m:n}, \delta_1, R_1), \ldots, (Y_{k:m:n}, \delta_k, R_k), \ldots, (Y_{d:m:n}, \delta_d, R_d), & \text{if } Y_{k:m:n} < T < Y_{m:m:n}, \\ \text{CaseIII}: & (Y_{1:m:n}, \delta_1, R_1), \ldots, (Y_{k:m:n}, \delta_k, R_k), \ldots, (Y_{m:m:n}, \delta_m, R_m), & \text{if } Y_{k:m:n} < Y_{m:m:n} < T. \end{cases} \tag{5}$$

Here $R_k = n - \sum_{i=1}^{k-1}(R_i + 1) - 1$ for Case I and d is the number of failures in Case II such that $Y_{d:m:n} < T < Y_{d+1:m:n}$. The notation δ_i is an indicator variable indicating whether failure cause for ith progressive censored time is detected or not, i.e.

$$\delta_i = \begin{cases} 0, & \text{if unit fails but the cause of failure is unknown,} \\ 1, & \text{if unit fails due to cause } c_i \text{ with } c_i = j,\ j = 1, 2. \end{cases}$$

Remark 2.1: Note that cause of failure under data (5) is partially observed and so we have the following situations.

(a) (y, c_i): test unit failed at time y and corresponding failure cause is c_i, that is $\delta_i = 1$, $c_i = j, j = 1, 2$.
(b) (y, δ^*): test unit failed at time y and corresponding failure cause is not known that is $\delta_i = 0$.
(c) $(y^*, *)$: test unit withdrawn at time y and is not failed till that time.

The likelihood contributions from these observations are as follows:

$$\text{(a) } h_j(y)\prod_{j=1}^{2} s_j(y), \quad \text{(b) } f(y), \ (c)\ S(y).$$

The joint density function of GPHC competing risk data (5) with partially observed causes of failures is given by

$$L(data) = \begin{cases} C_1 \prod_{i=1}^{k} [h_{c_i}(y_{i:m:n}) \prod_{j=1}^{2} s_j(y_{i:m:n})]^{\delta_i} [f(y_{i:m:n})]^{1-\delta_i} [S(y_{i:m:n})]^{R_i}, & \text{Case I,} \\ C_2 \prod_{i=1}^{d} [h_{c_i}(y_{i:m:n}) \prod_{j=1}^{2} s_j(y_{i:m:n})]^{\delta_i} [f(y_{i:m:n})]^{1-\delta_i} [S(y_{i:m:n})]^{R_i} [S(T)]^{R^*_{d+1}}, & \text{Case II,} \\ C_3 \prod_{i=1}^{m} [h_{c_i}(y_{i:m:n}) \prod_{j=1}^{2} s_j(y_{i:m:n})]^{\delta_i} [f(y_{i:m:n})]^{1-\delta_i} [S(y_{i:m:n})]^{R_i}, & \text{Case III,} \end{cases}$$

where $R^*_{d+1} = n - \sum_{i=1}^{d}(R_i + 1)$ and

$$\begin{cases} C_1 = \prod_{j=1}^{k} \sum_{s=j}^{m} (R_s + 1), & \text{Case I,} \\ C_2 = \prod_{j=1}^{d} \sum_{s=j}^{m} (R_s + 1), & \text{Case II,} \\ C_3 = \prod_{j=1}^{m} \sum_{s=j}^{m} (R_s + 1), & \text{Case III.} \end{cases}$$

Now suppose GPHC competing risk failure data (5) are drawn from a Kumaraswamy population with parameters α, λ_1 and λ_2. The likelihood function of α, λ_1 and λ_2 is now

obtained as

$$L(\alpha,\lambda_1,\lambda_2;\text{data}) = \begin{cases} C_1\prod_{i=1}^{k}\alpha\lambda_{c_i}^{\delta_i}\lambda_0^{1-\delta_i}y_{i:m:n}^{\alpha-1}(1-y_{i:m:n}^{\alpha})^{(R_i+1)\lambda_0-1}, & \text{Case I;} \\ C_2\prod_{i=1}^{d}\alpha\lambda_{c_i}^{\delta_i}\lambda_0^{1-\delta_i}y_{i:m:n}^{\alpha-1}(1-y_{i:m:n}^{\alpha})^{(R_i+1)\lambda_0-1}(1-T^{\alpha})^{\lambda_0 R^*_{d+1}}, & \text{Case II;} \\ C_3\prod_{i=1}^{m}\alpha\lambda_{c_i}^{\delta_i}\lambda_0^{1-\delta_i}y_{i:m:n}^{\alpha-1}(1-y_{i:m:n}^{\alpha})^{(R_i+1)\lambda_0-1}, & \text{Case III,} \end{cases} \quad (6)$$

where $\lambda_0 = \lambda_1 + \lambda_2$.

Remark 2.2: Suppose $0 < p < 1$ denotes masking probability when the competing risks data are partially observed. The problem turns to a usual completely observed competing risks problem when the masking probability takes the value $p = 0$ where as it turns out to be a general problem with no competing risks, i.e. the cause of failure for all the data observed is missing when the value of p is 1.

Remark 2.3: In this paper, the competing risks model is constructed on the basis of Cox's latent failure times model procedures but mixture model approach can also be considered as an alternatively to this model. In which it can be considered that the latent failures times follow mixture model of two Kumaraswamy distribution which is given as

$$F(x) = pF_1(x;\alpha,\lambda_1) + (1-p)F_2(x;\alpha,\lambda_2), \quad x > 0,$$

where $0 < p < 1$ is the unknown mixture parameter. Further, cause-specific hazard function approach can also be used for analysis of competing risk data where different hazard functions are assumed for cause of failures. One may see Cheng and Fine [3], Narendranathan and Stewart [29] for a detailed discussion.

2.2. Prior distributions

Here we discuss the prior distributions for unrestricted and order restricted parameters cases. We mention that the inference upon order restricted parameters has found wide applications in many practical field of study such as mortality analysis, clinical trials, bio-assays, reliability and life testing experiments. So along with unrestricted parameters, estimation of order restricted parameters is also quite important in practice.

2.2.1. Prior without order restriction

Here we discuss prior distributions under the case parameters λ_1 and λ_2 are unrestricted. Let $\lambda_0 = (\lambda_1 + \lambda_2)$ follow a gamma prior distribution with hyperparameters (a_0, b_0) (see

Pena and Gupta [30]) with density function

$$\Pi_0(\lambda_0) = \frac{b_0^{a_0}}{\Gamma(a_0)}\lambda_0^{a_0-1}e^{-b_0\lambda_0}, \quad a_0,\ b_0,\ \lambda_0 > 0.$$

Also assume that, for given λ_0, (λ_1/λ_0) follow a conditional beta prior distribution with hyper-parameters (p, q) with probability density

$$\Pi_0\left(\frac{\lambda_1}{\lambda_0}\middle|\lambda_0\right) = \frac{\Gamma(p+q)}{\Gamma(p)\Gamma(q)}\left(\frac{\lambda_1}{\lambda_0}\right)^{p-1}\left(\frac{\lambda_2}{\lambda_0}\right)^{q-1}.$$

Thus joint prior of λ_1 and λ_2, namely gamma-beta GB(a_0, b_0, p, q) distribution (see, Mondal and Kundu [24,25]), is obtained as

$$\Pi_0(\lambda_1,\lambda_2) = \frac{\Gamma(p+q)}{\Gamma(a_0)\Gamma(p)\Gamma(q)}\frac{b_0^{a_0}\lambda_1^{p-1}\lambda_2^{q-1}}{(\lambda_1+\lambda_2)^{p+q-a_0}}e^{-b_0(\lambda_1+\lambda_2)}, \quad \lambda_1,\lambda_2 > 0. \tag{7}$$

In addition, an independent prior for α is now considered as

$$\Pi(\alpha) = \frac{b_1^{a_1}}{\Gamma(a_1)}\alpha^{a_1-1}e^{-b_1\alpha}, \quad a_1,\ b_1 > 0,\ \alpha > 0. \tag{8}$$

2.2.2. Prior under order restriction

Under the restriction $\lambda_1 \le \lambda_2$, the corresponding joint prior distribution is given by

$$\Pi_0^*(\lambda_1,\lambda_2) = \frac{\Gamma(p+q)}{\Gamma(a_0)\Gamma(p)\Gamma(q)}\frac{b_0^{a_0}(\lambda_1^{p-1}\lambda_2^{q-1}+\lambda_1^{q-1}\lambda_2^{p-1})}{(\lambda_1+\lambda_2)^{p+q-a_0}}e^{-b_0(\lambda_1+\lambda_2)}, \quad 0 < \lambda_1 < \lambda_2 < \infty. \tag{9}$$

This is referred as ordered gamma-beta OGB(a_0, b_0, p, q) prior distribution (see, Mondal and Kundu [24,25]). The prior for α is considered as a gamma distribution as defined above.

3. Estimation without order restriction

In this section, we obtain point and interval estimators using classical and Bayesian approaches under the competing risks model when data are observed using GPHC. In this section parameters λ_1 and λ_2 are unrestricted.

3.1. Maximum likelihood estimation

From (6), the likelihood function can be re-written as

$$L(\alpha,\lambda_1,\lambda_2;\text{data}) = C^*\alpha^{d^*}\lambda_0^{e_0}\lambda_1^{e_1}\lambda_2^{e_2}e^{-\lambda_0\psi(\alpha)}\prod_{i=1}^{d^*}\frac{y_{i:m:n}^{\alpha-1}}{1-y_{i:m:n}^{\alpha}} \tag{10}$$

where d^* is the total failures under respective cases:

$$d^* = \begin{cases} k, & \text{Case I,} \\ d, & \text{Case II,} \\ m, & \text{Case III,} \end{cases} \tag{11}$$

and

$$\psi(\alpha) = \begin{cases} -\sum_{i=1}^{k}[(1+R_i)\ln(1-y_{i:m:n}^{\alpha})], & \text{Case I,} \\ -\sum_{i=1}^{d}[(1+R_i)\ln(1-y_{i:m:n}^{\alpha})] - R_{d+1}^{*}\ln(1-T^{\alpha}), & \text{Case II,} \\ -\sum_{i=1}^{m}[(1+R_i)\ln(1-y_{i:m:n}^{\alpha})], & \text{Case III,} \end{cases}$$

and C^*, e_0 and e_j are, respectively,

$$C^* = \prod_{j=1}^{d^*}\sum_{k=j}^{m}(R_k+1),$$

$$e_0 = \sum_{i=1}^{d^*}(1-\delta_i): \text{ total number of observed failure times with failure causes unknown}$$

$$e_j = \sum_{i=1}^{d^*}\delta_i \mathbf{1}_{(c_i=j)},\ j=1,2: \text{ number of failures under cause } j.$$

The corresponding log-likelihood function is given by

$$l(\alpha,\lambda_1,\lambda_2;\text{data}) = \ln C^* + d^*\ln\alpha + e_0\ln\lambda_0 + \sum_{j=1}^{2} e_j\ln\lambda_j - \lambda_0\psi(\alpha) + \sum_{i=1}^{d^*}\ln\left(\frac{y_{i:m:n}^{\alpha-1}}{1-y_{i:m:n}^{\alpha}}\right). \tag{12}$$

Theorem 3.1: *Let at least one failure be available due to cause j, that is, $e_j \geq 1, j = 1, 2$, then MLE of λ_j for a given α is obtained as*

$$\lambda_j = \frac{e_j}{\psi(\alpha)}\left[\frac{e_0+e_1+e_2}{e_1+e_2}\right], \quad j=1,2. \tag{13}$$

Proof: See Supplemental Material. ■

Remark 3.1: For existence of MLE of λ_j for given α, it is necessary to have at least one failure due to cause *j* for which both failure time and associated cause of failure are observed. It is also observed that the MLEs of λ_1 and λ_2 is proportional to e_1/e_2, i.e. the ratio of the number of failures due to cause one and cause two. Further, it can be concluded that if there is no failure due to cause *j*, i.e. $e_j = 0$, $j = 1, 2$, then the observed data provide no information about λ_j. Therefore, when $e_j = 0$ the associated MLE of λ_j does not exist.

Remark 3.2: Suppose $\lambda_1 = \lambda_2$, the likelihood function coincides with that of the GPHC data from Kumaraswamy distribution with parameters α and $\lambda_1 + \lambda_2$. Thus MLE of $\lambda_1 + \lambda_2$ for given α can be expressed as follows

$$\widetilde{\lambda_1+\lambda_2} = \frac{e_0+e_1+e_2}{\psi(\alpha)}.$$

Theorem 3.2: *Let at least one failure due to cause j, that is, $e_j \geq 1$, $j = 1, 2$, be available, then MLE of α exists uniquely which can be computed from the following equation*

$$\frac{1}{\alpha} + \frac{1}{d^*} \sum_{i=1}^{d^*} \left(\frac{\ln y_{i:m:n}}{1 - y_{i:m:n}^{\alpha}} \right) = \frac{\psi'(\alpha)}{\psi(\alpha)}. \tag{14}$$

Proof: See Supplemental Material. ■

Suppose the MLE of α is denoted by $\hat{\alpha}$ which can easily be obtained using the fixed point iteration algorithm from the equation (14). Then the MLEs of the parameters λ_1 and λ_2 denoted by $\hat{\lambda}_1$ and $\hat{\lambda}_2$, respectively, can be obtained from Theorem 3.1 as

$$\hat{\lambda}_1 = \frac{e_1}{\psi(\hat{\alpha})} \left[\frac{e_0 + e_1 + e_2}{e_1 + e_2} \right] \quad \text{and} \quad \hat{\lambda}_2 = \frac{e_2}{\psi(\hat{\alpha})} \left[\frac{e_0 + e_1 + e_2}{e_1 + e_2} \right].$$

3.2. Approximate confidence interval

Here confidence intervals for α, λ_1 and λ_2 are constructed using the asymptotic normality property of MLEs. Assume that $\theta = (\theta_1, \theta_2, \theta_3)$ with $\theta_1 = \lambda_1$, $\theta_2 = \lambda_2$ and $\theta_3 = \alpha$ then observed Fisher information matrix is given by

$$I(\theta) = \left[-\frac{\partial^2 l(\lambda_1, \lambda_2, \alpha; \text{data})}{\partial \theta_j \theta_s} \right]_{\theta = \hat{\theta}, j, s = 1,2,3} = [I_{js}]_{3 \times 3} \tag{15}$$

with elements

$$I_{11} = \frac{e_0}{\lambda_0^2} + \frac{e_1}{\lambda_1^2}, \quad I_{12} = I_{21} = \frac{e_0}{\lambda_0^2}, \quad I_{13} = I_{31} = I_{23} = I_{32} = \psi'(\alpha),$$

$$I_{22} = \frac{e_0}{\lambda_0^2} + \frac{e_2}{\lambda_2^2}, \quad I_{33} = \frac{d^*}{\alpha^2} + \lambda_0 \psi''(\alpha) + \sum_{i=1}^{d^*} \{ y_i^{\alpha} (\ln y_i)^2 / (1 - y_i^{\alpha})^2 \}.$$

where

$$\psi''(\alpha) = \begin{cases} \sum_{i=1}^{k} (1 + R_i)\, y_i^{\alpha} \{ \ln y_i / (1 - y_i^{\alpha}) \}^2, & \text{Case I,} \\ \sum_{i=1}^{d} (1 + R_i)\, y_i^{\alpha} \{ \ln y_i / (1 - y_i^{\alpha}) \}^2 + R_{d+1}^{*} T^{\alpha} \{ \ln T / (1 - T^{\alpha}) \}^2, & \text{Case II,} \\ \sum_{i=1}^{m} (1 + R_i)\, y_i^{\alpha} \{ \ln y_i / (1 - y_i^{\alpha}) \}^2, & \text{Case III.} \end{cases}$$

The asymptotic distribution of MLE $\hat{\theta}$ is such that $\hat{\theta} - \theta$ follows $N(0, I^{-1}(\theta))$ distribution. Now inverse of the information matrix is

$$I^{-1}(\theta) = \begin{pmatrix} \text{Var}(\hat{\lambda}_1) & \text{Cov}(\hat{\lambda}_1, \hat{\lambda}_2) & \text{Cov}(\hat{\lambda}_1, \hat{\alpha}) \\ \text{Cov}(\hat{\lambda}_2, \hat{\lambda}_1) & \text{Var}(\hat{\lambda}_2) & \text{Cov}(\hat{\lambda}_2, \hat{\alpha}) \\ \text{Cov}(\hat{\alpha}, \hat{\lambda}_1) & \text{Cov}(\hat{\alpha}, \hat{\lambda}_2) & \text{Var}(\hat{\alpha}) \end{pmatrix}.$$

The $100(1-\epsilon)\%$, $0 < \epsilon < 1$, approximate confidence interval (ACI) of θ_j is now given by $(\hat{\theta}_j \pm z_{\epsilon/2}\sqrt{\mathrm{Var}(\hat{\theta}_j)})$, $j = 1, 2, 3$ with $z_{\epsilon/2}$ is the upper $\frac{\epsilon}{2}$th quantile of the standard normal distribution.

3.3. Bayesian estimation

In this section, Bayes estimates of unknown parameters are obtained under competing risks situations when λ_1 and λ_2 are unrestricted and data are observed using generalized progressive hybrid censoring. The joint posterior distribution of α, λ_1 and λ_2 is of the following form:

$$\Pi(\alpha, \lambda_1, \lambda_2 \mid \text{data}) = \frac{C^*\Gamma(p+q)}{\Gamma(a_0)\Gamma(a_1)\Gamma(p)\Gamma(q)} \frac{b_0^{a_0} b_1^{a_1} \lambda_1^{p+e_1-1} \lambda_2^{q+e_2-1}}{(\lambda_1+\lambda_2)^{p+q-a_0-e_0}} e^{-b_0(\lambda_1+\lambda_2)} \alpha^{a_1+d^*-1} \times e^{-b_1\alpha} e^{-(\lambda_1+\lambda_2)\psi(\alpha)} \prod_{i=1}^{d^*} \frac{y_i^{\alpha-1}}{1-y_i^{\alpha}}. \tag{16}$$

This is of the type

$$\Pi(\alpha, \lambda_1, \lambda_2 \mid \text{data}) = \Pi(\alpha \mid \text{data}) \times \Pi(\lambda_1, \lambda_2 \mid \alpha, \text{data}), \tag{17}$$

where

$$\Pi(\alpha \mid \text{data}) \propto \frac{\alpha^{a_1+d^*-1}\, e^{-b_1\alpha + \sum_{i=1}^{d^*} \ln\left(\frac{y_i^{\alpha-1}}{1-y_i^{\alpha}}\right)}}{[b_0+\psi(\alpha)]^{(a_0+d^*)}} \tag{18}$$

and

$$\Pi(\lambda_1, \lambda_2 \mid \alpha, \text{data}) \propto \frac{[b_0+\psi(\alpha)]^{(a_0+d^*)}\, \lambda_1^{[p+e_1]-1} \lambda_2^{[q+e_2]-1}}{(\lambda_1+\lambda_2)^{[p+e_1]+[q+e_2]-a_0-d^*}} e^{-[b_0+\psi(\alpha)](\lambda_1+\lambda_2)} = BG(a_0+d^*,\, b_0+\psi(\alpha),\, p+e_1,\, q+e_2). \tag{19}$$

Since it is difficult to generate random samples from $\Pi(\alpha \mid \text{data})$. However from the profile plots of $\Pi(\alpha \mid \text{data})$ it is observed that this distribution may be uni-model. So we approximate it from a gamma distribution with parameters ς and τ which are derived by equating mean and variance of $\Pi(\alpha \mid \text{data})$ to those of the gamma distribution (see Ganguly *et al.* [10]). Let μ and σ be mean and variance of $\Pi(\alpha \mid \text{data})$, then shape and scale parameters of approximating gamma distribution are obtained as $\varsigma = \frac{\mu^2}{\sigma}$ and $\tau = \frac{\mu}{\sigma}$, respectively. The probability distribution of approximating gamma distribution then turns out to be

$$\pi(\alpha \mid \text{data}) = \frac{\tau^{\varsigma}}{\Gamma(\varsigma)} \alpha^{\varsigma-1} e^{-\tau\alpha}, \quad \alpha > 0.$$

Thus, $\Pi(\alpha, \lambda_1, \lambda_2 \mid \text{data})$ can be rewritten as

$$\Pi(\alpha, \lambda_1, \lambda_2 \mid \text{data}) = h(\alpha) \times \pi(\alpha \mid \text{data}) \times \Pi(\lambda_1, \lambda_2 \mid \text{data}),$$

where $h(\alpha) = \frac{\Pi(\alpha|\text{data})}{\pi(\alpha|\text{data})}$.

To compute Bayesian estimates of arbitrary parametric function $\phi(\alpha, \lambda_1, \lambda_2)$ of $\alpha, \lambda_1, \lambda_2$, the following algorithm is proposed.

Bayesian sampling algorithm under non-restriction case

Step 1 Generate α from $Gamma(\varsigma, \tau)$ distribution.

Step 2 For given α, generate λ_1 and λ_2 from $BG(a_0 + d^*,\ b_0 + \psi(\alpha),\ p + e_1,\ q + e_2)$, (Kundu and Pradhan [15]).

Step 3 Repeat Steps 1-2, N times to simulate $\{(\alpha_1, \lambda_{11}, \lambda_{21}), \ldots, (\alpha_N, \lambda_{1N}, \lambda_{2N})\}$.

Step 4 Compute $\phi_i = \phi(\alpha_i, \lambda_{1i}, \lambda_{2i});\ i = 1, 2, \ldots, N$.

Step 5 Calculate $h_i = h(\alpha_i),\ i = 1, 2, \ldots, N$.

Step 6 Find Bayes estimate of $\phi(\alpha, \lambda_1, \lambda_2)$ as $\hat{\phi}(\alpha, \lambda_1, \lambda_2) = \frac{1}{N}\sum_{i=1}^{N} h_i\phi_i$.

Step 7 For $100(1-\epsilon)\%$ symmetric credible interval, first order ϕ_i for $i = 1, 2, \ldots, N$, say, $\phi_{(1)} < \phi_{(2)} < \cdots < \phi_{(N)}$ and order h_i accordingly to get $h_{(1)}, h_{(2)}, \ldots, h_{(N)}$. Here $h_{(1)}, h_{(2)}, \ldots, h_{(N)}$ may not be ordered. Then $100(1-\epsilon)\%$ symmetric interval is given by $(\phi_{(i_1)}, \phi_{(i_2)})$, where i_1 and i_2 are such that

$$i_1, i_2 \in \{1, 2, \ldots, N\},\ i_1 < i_2, \quad \frac{1}{N}\sum_{j=i_1}^{i_2} h_{(j)} \le 1-\epsilon \le \frac{1}{N}\sum_{j=i_1}^{i_2+1} h_{(j)}. \tag{20}$$

The $100(1-\epsilon)\%$ HPD credible interval of $\phi(\alpha, \lambda_1, \lambda_2)$ can be constructed as $(\phi_{(i_1^*)}, \phi_{(i_2^*)})$, where $i_1^* < i_2^*, i_1^*, i_2^* \in \{1, 2, \ldots, N\}$ satisfy

$$\frac{1}{N}\sum_{j=i_1^*}^{i_2^*} h_{(j)} \le 1-\epsilon \le \frac{1}{N}\sum_{j=i_1^*}^{i_2^*+1} h_{(j)}, \quad \phi(i_2^*) - \phi(i_1^*) \le \phi(i_2) - \phi(i_1),$$

for all i_1 and i_2 satisfying (20).

Notice that when all hyper-parameters are zero, the priors refer to non-informative priors. Under this case, the posterior density functions for α and (λ_1, λ_2) are still proper. The corresponding Bayes estimates under the non-informative prior are obtained similarly using the above algorithm.

4. Estimation under order restriction

Here unknown parameters of Kumaraswamy distributions are estimated under the restriction $\lambda_1 \le \lambda_2$ based on GPHC. Both classical and Bayesian approaches are discussed from competing risks viewpoint.

4.1. Classical estimation

Theorem 4.1: *Let at least one failure due to cause j, that is $j \ge 1$, $j = 1, 2$ be available. For a given α, MLE $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$ of λ_1 and λ_2 are obtained as,*

$$\begin{cases} \tilde{\lambda}_1 = \dfrac{e_j}{\psi(\alpha)}\left[\dfrac{e_0 + e_1 + e_2}{e_1 + e_2}\right],\ j = 1, 2, & \text{if } e_1 < e_2; \\ \tilde{\lambda}_1 = \tilde{\lambda}_2 = \dfrac{d^*}{2\psi(\alpha)}, & \text{if } e_1 \ge e_2. \end{cases}$$

Proof: See Supplemental Material. ■

Theorem 4.2: *Let at least one failure due to cause j, that is* $j \geq 1$, $j = 1, 2$ *be available. Then MLE of* α *uniquely exits which is same as the estimate presented in the Theorem 3.2.*

Proof: Considering $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$ in the log-likelihood (12) for cases $e_1 < e_2$ and $e_1 \geq e_2$, respectively, it is seen that the profile log-likelihood function of α by ignoring constant has the same expression as given Theorem 3.2. Thus MLE of α under order restriction $\lambda_1 \leq \lambda_2$ can be computed from the likelihood Equation (14). This proves the theorem. ■

Now from Theorems 4.1 and 4.2, the MLEs of parameters α, λ_1 and λ_2, denoted as $\tilde{\alpha}$, $\tilde{\lambda}_1$, $\tilde{\lambda}_2$, respectively, can be obtained as

$$\tilde{\alpha} = \hat{\alpha} \text{and} \begin{cases} \tilde{\lambda}_j = \dfrac{e_j}{\psi(\tilde{\alpha})}\left[\dfrac{e_0 + e_1 + e_2}{e_1 + e_2}\right], \; j = 1, 2, & \text{if } e_1 < e_2; \\ \tilde{\lambda}_1 = \tilde{\lambda}_2 = \dfrac{d^*}{2\psi(\tilde{\alpha})}, & \text{if } e_1 \geq e_2. \end{cases}$$

The ACIs of these parameters can be constructed by direct substitution of MLEs, obtained under order restriction, in the expression given in Section 3.2.

4.2. Estimation under bayes procedure

Following Section 3, the joint posterior distribution of α, λ_1 and λ_2 using Equations (8), (9) and (10) is given by

$$\Pi^*(\alpha, \lambda_1, \lambda_2) = \Pi^*(\alpha \mid \text{data}) \times \Pi^*(\lambda_1, \lambda_2 \mid \alpha, \text{data}) \times \eta(\alpha, \lambda_1, \lambda_2) \tag{21}$$

where $e_3 = \min(e_1, e_2)$, $\eta(\alpha, \lambda_1, \lambda_2) = \lambda_1^{e_1 - e_3} \lambda_2^{e_2 - e_3}$,

$$\Pi^*(\alpha \mid \text{data}) \propto \frac{\alpha^{a_1 + d^* - 1} \, e^{-b_1 \alpha + \sum_{i=1}^{d^*} \ln\left(\frac{y_i^{\alpha - 1}}{1 - y_i^{\alpha}}\right)}}{[a_0 + \psi(\alpha)]^{a_0 + e_0 + 2e_3}} \tag{22}$$

and

$$\Pi^*(\lambda_1,\lambda_2 \mid \alpha, \text{data}) \propto \frac{[b_0+\psi(\alpha)]^{a_0+e_0+2e_3}\left(\lambda_1^{p+e_3-1}\lambda_2^{q+e_3-1}+\lambda_1^{q+e_3-1}\lambda_2^{p+e_3-1}\right)}{(\lambda_1+\lambda_2)^{[p+e_3]+[q+e_3]-a_0-e_0-2e_3}} \times e^{-[b_0+\psi(\alpha)](\lambda_1+\lambda_2)}$$
$$= OGB(a_0+e_0+2e_3, b_0+\psi(\alpha), p+e_3, q+e_3).$$

Here again, we approximate $\Pi^*(\lambda \mid \text{data})$ by a gamma distribution with parameters ς^* and τ^* which are derived by equating the mean and variance of $\Pi^*(\alpha \mid \text{data})$ to those of the gamma distribution. Suppose μ^* and σ^* denote the mean and variance of $\Pi^*(\alpha \mid \text{data})$, respectively, then following the similar procedure as in Section 3, the obtained shape and scale parameters of the approximating gamma distribution are $\varsigma^* = \frac{\mu^{*2}}{\sigma^*}$ and $\tau^* = \frac{\mu^*}{\sigma^*}$, respectively. The corresponding density is given by

$$\pi^*(\alpha \mid \text{data}) = \frac{\tau^{*\varsigma^*}}{\Gamma(\varsigma^*)}\alpha^{\varsigma^*-1}e^{-\tau^*\alpha}, \quad \alpha > 0.$$

Therefore,

$$\Pi^*(\alpha,\lambda_1,\lambda_2 \mid \text{data}) = \eta^*(\alpha,\lambda_1,\lambda_2) \times \pi^*(\alpha \mid \text{data}) \times \Pi(\lambda_1,\lambda_1 \mid \alpha, \text{data})$$

where $\eta^* = h^*(\alpha)\eta(\alpha,\lambda_1,\lambda_2)$ and $h^*(\alpha) = \frac{\Pi^*(\alpha|\text{data})}{\pi^*(\alpha|\text{data})}$. From the above distribution Bayes point and interval estimates of $\phi(\alpha,\lambda_1,\lambda_2)$ can be computed under the order restricted case. The following algorithm is proposed based on sample generated using importance sampling technique.

Bayeisian sampling algorithm under parameter restriction case

Step 1 Generate α from $Gamma(\varsigma^*, \tau^*)$ distribution.

Step 2 For given α, generate λ_1 and λ_2 from $OGB(a_0+e_1+2e_3,\ b_0+\psi(\alpha),\ p+e_3,\ q+e_3)$ distribution.

Step 3 Repeat Steps 1-2, N times to simulate $\{(\alpha_1,\lambda_{11},\lambda_{21}),\ldots,(\alpha_N,\lambda_{1N},\lambda_{2N})\}$.

Step 4 Compute $\phi_i = \phi(\alpha_i,\lambda_{1i},\lambda_{2i});\ i = 1,2,\ldots,N$.

Step 5 Calculate $\eta_i^* = \eta^*(\alpha_i,\lambda_{1i},\lambda_{2i}),\ i = 1,2,\ldots,N$.

Step 6 Find Bayes estimate of $\phi(\alpha,\lambda_1,\lambda_2)$ as

$$\sum_{i=1}^{N}\eta_i^{**}\phi_i, \quad \text{where } \eta_i^{**} = \frac{\eta_i^*}{\sum_{i=1}^N \eta_i^*}.$$

Step 7 To construct $100(1-\epsilon)\%$ symmetric credible interval, order $\phi_i, i = 1,2,\ldots,N$, as $\phi_{(1)} < \phi_{(2)} < \cdots < \phi_{(N)}$ and order η_i^* correspondingly to obtain $\eta_{(1)}^{**}, \eta_{(2)}^{**}, \ldots, \eta_{(N)}^{**}$. Here $\eta_{(1)}^{**}, \eta_{(2)}^{**}, \ldots, \eta_{(N)}^{**}$ may not be ordered. Then, $100(1-\epsilon)\%$ symmetric interval is given by $(\phi_{(i_1)}, \phi_{(i_2)})$, where i_1 and i_2 are such that

$$i_1, i_2 \in \{1,2,\ldots,N\},\ i_1 < i_2, \quad \sum_{j=i_1}^{i_2}\eta_{(j)}^{**} \le 1-\alpha \le \sum_{j=i_1}^{i_2+1}\eta_{(j)}^{**}. \tag{23}$$

Subsequently $100(1-\epsilon)\%$ HPD credible interval of $\phi(\alpha,\lambda_1,\lambda_2)$ is given by $(\phi_{(i_1^*)},\phi_{(i_2^*)})$, where $i_1^* < i_2^*, i_1^*, i_2^* \in \{1,2,\ldots,N\}$ satisfy

$$\sum_{j=i_1^*}^{i_2^*} \eta_{(j)}^{**} \le 1-\epsilon \le \sum_{j=i_1^*}^{i_2^*+1} \eta_{(j)}^{**},\ \phi(i_2^*)-\phi(i_1^*) \le \phi(i_2)-\phi(i_1),$$

for all i_1 and i_2 satisfying (23).

5. Testing problem

In previous sections, we obtained various inferences on the basis of partially observed competing risks model when failures due to cause 1 and cause 2 follow Kumaraswamy distributions with common shape parameter $\alpha_1=\alpha_2=\alpha$ and unequal shape parameters λ_1 and λ_2, respectively. We now verify equivalence of shape parameters α_1 and α_2 for illustration purposes. We propose the following hypothesis testing problem:

$$H_0 : \alpha_1=\alpha_2=\alpha \quad \text{versus} \quad H_1 : \alpha_1 \neq \alpha_2.$$

Suppose $l_1(\hat{\alpha},\hat{\lambda}_1,\hat{\lambda}_2)$ denotes the log-likelihood with common shape parameter α and $l_2(\hat{\alpha}_1,\hat{\alpha}_2,\hat{\lambda}_1,\hat{\lambda}_2)$ be the log-likelihood function with unequal shape parameters α_1 and α_2, respectively. The likelihood ratio statistic is now given as

$$LRS = -2(l_1(\hat{\alpha},\hat{\lambda}_1,\hat{\lambda}_2) - l_2(\hat{\alpha}_1,\hat{\alpha}_2,\hat{\lambda}_1,\hat{\lambda}_2)) \to \chi_1^2.$$

The associated test can be constructed using the asymptotic distribution of *LRS*. We reject H_0 if $LRS > c^*$ where c^* is such that $P(\chi_1^2 > c^*)$= size of the test (ϵ).

6. Numerical illustrations

6.1. Simulation study

In this section, we perform extensive Monte Carlo simulations to compare the performance of proposed estimation methods under generalized progressive hybrid censoring. The point estimators of competing risks model parameters are compared with respect to their estimated mean squared errors (MSEs). Interval estimators are examined on the basis of average interval widths (AIWs) and associated coverage probabilities (CPs). The actual values of parameters are arbitrarily taken as $(\alpha,\lambda_1,\lambda_2)=(1.5,0.7,0.9)$. We perform this simulation study by considering $(n,m)=(60,50)$ and $(80,70)$ under different combinations of k, masking probability u and three different censoring schemes described below:

CS-1: $R_1=R_2=\cdots=R_{m-1}=0$ and $R_m=n-m$;
CS-2: $R_1=n-m$ and $R_2=R_3=\cdots=R_m=0$;
CS-3: $R_1=R_2=\cdots=R_{n-m}=1$ and $R_{n-m+1}=\cdots=R_m=0$.

Here CS-1 corresponds to the traditional type-II censoring scheme and CS-2 is the first step-censoring. The scheme CS-3 is accordingly described. For the first set $(n,m)=$

(60, 50), the criteria values are obtained for $k = 40$ and 45 under different masking probabilities such as $p = 0.05$ and 0.10. Similarly for $(n, m) = (80, 70)$, the criteria values are obtained for $k = 60$ and 65 under the same masking probabilities and censoring schemes. All these results are obtained for two different values of $T = 0.5$ and 0.6. We compute Bayes estimates against informative prior assumption with hyper-parameters $a_0 = 3.2$, $b_0 = 2$, $p = 7/9$, $q = 1$, $a_1 = 6$ and $b_1 = 4$. The hyperparameters of the prior distributions for each of the parameters are chosen in such a way that the prior means of corresponding parameters are relatively close to true values of the parameters. Under this framework, we first generate partially observed competing risks GPHC samples. The corresponding algorithm is given below:

Algorithm for Sample Generation:

Step 1: Generate progressive type-II censored samples from Kumaraswamy distribution with parameters α and $(\lambda_1 + \lambda_2)$.

Step 2: Assign failure causes to each of PCS type-II sample as 1 or 2 with probability $\frac{\lambda_1}{\lambda_1 + \lambda_2}$ and $\frac{\lambda_2}{\lambda_1 + \lambda_2}$, respectively.

Step 3: Compute competing risks GPHC samples by comparing T, $Y_{k:m:n}$ and the PCS type-II sample generated in Step 2.

Step 4: Obtain partially observed competing risks GPHC data by generating a trial of 0 and 1 of size d^* using a Bernoulli distribution with parameter p, where $0 < p < 1$ is the masking probability. The associated failure cause is considered as unknown if zero appears. The associated failure cause is considered as known if one appears in the Bernoulli trial.

All the point and interval estimates are obtained based on 10,000 repetitions. In Tables 1–4, the average point estimates (AEs) together with associated MSEs (listed below each AEs) are presented for both the cases when λ_1 and λ_2 are unrestricted (Tables 1 and 3) as well order restricted case (Tables 2 and 4). In Tables 5–8, the AIWs of asymptotic confidence intervals, Bayesian symmetric credible intervals and highest posterior density intervals are presented. The CPs are presented just below each AIWs for both the cases without order restriction (Tables 5 and 7) and order restriction (Tables 6 and 8) on λ_1 and λ_2. We draw following conclusions from the various results obtained in Tables 1–8:

(1) The MSEs of MLEs of unknown parameters α, λ_1 and λ_2 decrease with an increase in k and T values.
(2) For fixed n, m, k, and T, with an increase in the masking probability the MSEs of proposed estimates tend to increase.
(3) For fixed n, m, k, $p$and$T$, the MSEs of respective estimate of estimates under different censoring schemes are highly satisfactory. We observe that CS-2 and CS-3 marginally produce better results than CS-1 in most of the cases.
(4) For fixed sample size, Bayesian estimates have smaller MSEs compared to respective ML estimates.
(5) The AIWs of all the intervals tend to decrease with an increase in k and T values. The corresponding CPs are found to be quite satisfactory and remain close to the nominal level in most of the cases.
(6) For fixed n, m, k, and T, with an increase in the masking probability, the AIWs increase

Table 1. AEs and MSEs for the parameters α, λ_1 and λ_2 without order restriction.

$T = 0.5$				MLE			Bayesian estimation		
(n, m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	1.5754	0.7564	1.0439	1.5273	0.7171	0.9868
				0.0923	0.0627	0.1272	0.0434	0.0254	0.0507
			CS-2	1.5722	0.7362	1.0069	1.5342	0.7078	0.9659
				0.0834	0.0515	0.0955	0.0429	0.0259	0.0457
			CS-3	1.5729	0.7402	1.0166	1.5324	0.7098	0.9721
				0.0819	0.0500	0.1025	0.0422	0.0234	0.0457
		45	CS-1	1.5676	0.7339	1.0120	1.5299	0.7077	0.9729
				0.0886	0.0443	0.0936	0.0459	0.0209	0.0437
			CS-2	1.5685	0.7282	1.0112	1.5349	0.7050	0.9758
				0.0792	0.0400	0.0841	0.0439	0.0201	0.0451
			CS-3	1.5578	0.7232	0.9987	1.5269	0.7021	0.9665
				0.0736	0.0376	0.0776	0.0426	0.0207	0.0400
	0.10	40	CS-1	1.5833	0.7473	1.0312	1.5353	0.7103	0.9766
				0.0974	0.0632	0.1281	0.0437	0.0258	0.0508
			CS-2	1.5794	0.7429	1.0258	1.5372	0.7119	0.9803
				0.0866	0.0515	0.1007	0.0443	0.0260	0.0476
			CS-3	1.5815	0.7503	1.0340	1.5365	0.7164	0.9845
				0.0897	0.0564	0.1060	0.0454	0.0265	0.0480
		45	CS-1	1.5631	0.7332	1.0169	1.5259	0.7067	0.9770
				0.0942	0.0434	0.0942	0.0459	0.0216	0.0443
			CS-2	1.5530	0.7203	0.9915	1.5238	0.7017	0.9631
				0.0792	0.0435	0.0847	0.0439	0.0209	0.0454
			CS-3	1.5581	0.7203	0.9982	1.5271	0.6995	0.9663
				0.0800	0.0383	0.0788	0.0463	0.0210	0.0401
(80,70)	0.05	60	CS-1	1.5726	0.7405	0.9915	1.5433	0.7214	0.9651
				0.0597	0.0354	0.0624	0.0357	0.0197	0.0362
			CS-2	1.5500	0.7266	0.9711	1.5267	0.7125	0.9509
				0.0568	0.0264	0.0494	0.0360	0.0166	0.0305
			CS-3	1.5542	0.7345	0.9812	1.5295	0.7196	0.9599
				0.0534	0.0272	0.0482	0.0338	0.0158	0.0298
		65	CS-1	1.5523	0.7298	0.9779	1.5269	0.7143	0.9561
				0.0594	0.0288	0.0554	0.0377	0.0176	0.0336
			CS-2	1.5400	0.7206	0.9600	1.5191	0.7089	0.9436
				0.0504	0.0217	0.0381	0.0328	0.0146	0.0249
			CS-3	1.5481	0.7278	0.9725	1.5250	0.7145	0.9535
				0.0521	0.0227	0.0442	0.0337	0.0146	0.0279
	0.10	60	CS-1	1.5617	0.7400	0.9856	1.5325	0.7210	0.9592
				0.0640	0.0358	0.0643	0.0380	0.0207	0.0374
			CS-2	1.5453	0.7194	0.9608	1.5238	0.7064	0.9424
				0.0569	0.0269	0.0495	0.0365	0.0171	0.0309
			CS-3	1.5428	0.7266	0.9660	1.5204	0.7131	0.9460
				0.0554	0.0275	0.0528	0.0359	0.0174	0.0320
		65	CS-1	1.5470	0.7307	0.9703	1.5236	0.7166	0.9509
				0.0597	0.0289	0.0569	0.0386	0.0179	0.0338
			CS-2	1.5566	0.7261	0.9650	1.5361	0.7146	0.9485
				0.0519	0.0217	0.0392	0.0355	0.0152	0.0264
			CS-3	1.5425	0.7248	0.9640	1.5222	0.7129	0.9472
				0.0533	0.0227	0.0443	0.0338	0.0154	0.0283

(7) The AIWs of different interval estimates under given censoring schemes are quite satisfactory.

(8) It is also observed that better estimates are obtained under order restriction compared to unrestricted case. In other words, additional information of order restriction about the parameters lead to better estimates.

Table 2. AEs and MSEs for the parameters α, λ_1 and λ_2 with order restriction.

$T = 0.5$				MLE			Bayesian estimation		
m	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	1.5754	0.7529	1.0412	1.5146	0.6825	0.9635
				0.0923	0.0617	0.1228	0.0457	0.0245	0.0482
			CS-2	1.5722	0.7289	1.0008	1.5224	0.6742	0.9456
				0.0834	0.0471	0.0942	0.0442	0.0231	0.0423
			CS-3	1.5729	0.7356	1.0128	1.5248	0.6800	0.9548
				0.0819	0.0481	0.1024	0.0444	0.0222	0.0448
		45	CS-1	1.5676	0.7338	1.0119	1.5200	0.6786	0.9529
				0.0886	0.0443	0.0936	0.0480	0.0206	0.0427
			CS-2	1.5685	0.7281	1.0111	1.5271	0.6802	0.9589
				0.0792	0.0399	0.0891	0.0446	0.0199	0.0428
			CS-3	1.5578	0.7226	0.9982	1.5225	0.6764	0.9556
				0.0736	0.0374	0.0776	0.0441	0.0194	0.0414
	0.10	40	CS-1	1.5833	0.7384	1.0241	1.5232	0.6743	0.9542
				0.0934	0.0619	0.1229	0.0458	0.0245	0.0489
			CS-2	1.5794	0.7313	1.0172	1.5283	0.6762	0.9607
				0.0866	0.0475	0.0987	0.0452	0.0234	0.0431
			CS-3	1.5815	0.7438	1.0297	1.5262	0.6794	0.9620
				0.0897	0.0520	0.1044	0.0472	0.0238	0.0455
		45	CS-1	1.5631	0.7321	1.0160	1.5148	0.6786	0.9589
				0.0892	0.0444	0.0937	0.0497	0.0211	0.0427
			CS-2	1.5530	0.7204	0.9919	1.5155	0.6716	0.9498
				0.0792	0.0400	0.0906	0.0461	0.0199	0.0430
			CS-3	1.5581	0.7182	0.9966	1.5203	0.6717	0.9534
				0.0800	0.0379	0.0776	0.0473	0.0200	0.0416
(80,70)	0.05	60	CS-1	1.5726	0.7397	0.9908	1.5305	0.6965	0.9391
				0.0597	0.0331	0.0623	0.0366	0.0178	0.0317
			CS-2	1.5500	0.7260	0.9705	1.5177	0.6898	0.9321
				0.0568	0.0258	0.0490	0.0383	0.0163	0.0297
			CS-3	1.5542	0.7346	0.9815	1.5220	0.6970	0.9430
				0.0534	0.0249	0.0478	0.0345	0.0152	0.0294
		65	CS-1	1.5523	0.7298	0.9779	1.5187	0.6938	0.9366
				0.0594	0.0288	0.0554	0.0394	0.0174	0.0330
			CS-2	1.5400	0.7207	0.9601	1.5131	0.6900	0.9282
				0.0504	0.0217	0.0380	0.0345	0.0142	0.0244
			CS-3	1.5481	0.7278	0.9725	1.5180	0.6946	0.9374
				0.0521	0.0227	0.0442	0.0354	0.0144	0.0272
	0.10	60	CS-1	1.5617	0.7354	0.9814	1.5178	0.6933	0.9355
				0.0640	0.0333	0.0642	0.0395	0.0194	0.0346
			CS-2	1.5453	0.7171	0.9591	1.5132	0.6815	0.9225
				0.0569	0.0259	0.0491	0.0386	0.0163	0.0299
			CS-3	1.5428	0.7248	0.9648	1.5124	0.6876	0.9288
				0.0554	0.0261	0.0524	0.0390	0.0178	0.0318
		65	CS-1	1.5470	0.7307	0.9704	1.5166	0.6948	0.9355
				0.0595	0.0289	0.0589	0.0395	0.0179	0.0337
			CS-2	1.5566	0.7257	0.9646	1.5259	0.6927	0.9331
				0.0519	0.0218	0.0392	0.0348	0.0144	0.0262
			CS-3	1.5425	0.7252	0.9645	1.5178	0.6913	0.9373
				0.0503	0.0228	0.0442	0.0369	0.0156	0.0288

(9) Estimates under $(n.m) = (80, 70)$ are generally better than those obtained using $(n.m) = (60, 50)$.

Thus we see that with an increase in k and T values, performance of all point and interval estimates improve with effective sample size increases. Further, it is observed

Table 3. AEs and MSEs for the parameters α, λ_1 and λ_2 without order restriction.

$T = 0.6$				MLE			Bayesian estimation		
(n, m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	1.5621	0.7337	1.0192	1.5209	0.7038	0.9748
				0.0911	0.0494	0.1018	0.0458	0.0236	0.0467
			CS-2	1.5802	0.7445	1.0371	1.5363	0.7115	0.9866
				0.0958	0.0533	0.1217	0.0483	0.0245	0.0521
			CS-3	1.5724	0.7407	1.0181	1.5312	0.7099	0.9728
				0.0901	0.0485	0.1014	0.0467	0.0234	0.0474
		45	CS-1	1.5640	0.7358	1.0123	1.5260	0.7089	0.9721
				0.0785	0.0454	0.0938	0.0407	0.0219	0.0438
			CS-2	1.5670	0.7351	1.0160	1.5316	0.7114	0.9804
				0.0800	0.0371	0.0829	0.0452	0.0195	0.0431
			CS-3	1.5703	0.7227	0.9967	1.5370	0.7007	0.9637
				0.0764	0.0374	0.0744	0.0433	0.0202	0.0392
	0.1	40	CS-1	1.5721	0.7330	1.0120	1.5288	0.7032	0.9678
				0.0922	0.0516	0.0984	0.0446	0.0234	0.0448
			CS-2	1.5791	0.7419	1.0267	1.5388	0.7126	0.9826
				0.0838	0.0495	0.0980	0.0433	0.0241	0.0460
			CS-3	1.5802	0.7472	1.0275	1.5370	0.7142	0.9789
				0.0933	0.0551	0.1119	0.0474	0.0263	0.0496
		45	CS-1	1.5911	0.7557	1.0341	1.5452	0.7224	0.9862
				0.0899	0.0506	0.1035	0.0442	0.0231	0.0444
			CS-2	1.5804	0.7378	1.0124	1.5439	0.7123	0.9751
				0.0820	0.0436	0.0860	0.0442	0.0231	0.0447
			CS-3	1.5574	0.7227	0.9925	1.5279	0.7030	0.9622
				0.0684	0.0348	0.0781	0.0390	0.0201	0.0422
(80,70)	0.05	60	CS-1	1.5624	0.7423	0.9866	1.5342	0.7231	0.9602
				0.0621	0.0356	0.0644	0.0379	0.0205	0.0369
			CS-2	1.5517	0.7290	0.9724	1.5292	0.7154	0.9530
				0.0547	0.0248	0.0475	0.0354	0.0160	0.0300
			CS-3	1.5458	0.7336	0.9790	1.5222	0.7182	0.9576
				0.0517	0.0292	0.0529	0.0330	0.0179	0.0320
		65	CS-1	1.5520	0.7313	0.9775	1.5282	0.7166	0.9567
				0.0552	0.0279	0.0530	0.0350	0.0171	0.0323
			CS-2	1.5397	0.7179	0.9584	1.5223	0.7081	0.9441
				0.0518	0.0202	0.0386	0.0351	0.0138	0.0260
			CS-3	1.5506	0.7269	0.9628	1.5285	0.7148	0.9459
				0.0473	0.0208	0.0363	0.0312	0.0138	0.0237
	0.1	60	CS-1	1.5558	0.7330	0.9802	1.5290	0.7160	0.9557
				0.0622	0.0361	0.0650	0.0380	0.0206	0.0370
			CS-2	1.5449	0.7220	0.9583	1.5244	0.7096	0.9411
				0.0548	0.0271	0.0475	0.0355	0.0182	0.0326
			CS-3	1.5428	0.7238	0.9738	1.5221	0.7108	0.9550
				0.0561	0.0299	0.0532	0.0362	0.0182	0.0327
		65	CS-1	1.5442	0.7323	0.9786	1.5198	0.7178	0.9578
				0.0576	0.0285	0.0548	0.0364	0.0177	0.0332
			CS-2	1.5444	0.7209	0.9597	1.5234	0.7093	0.9430
				0.0544	0.0229	0.0413	0.0352	0.0153	0.0268
			CS-3	1.5456	0.7235	0.9632	1.5239	0.7114	0.9456
				0.0503	0.0228	0.0431	0.0332	0.0150	0.0281

that with increase in masking probability, the performance of estimates becomes poor in terms of MSEs and AIWs. The ordered restricted estimates show better results than without order restriction. Finally large values of (n, m) provide better estimates of unknown parameters.

Table 4. AEs and MSEs for the parameters α, λ_1 and λ_2 with order restriction.

$T = 0.6$				MLE			Bayesian estimation		
(n, m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	1.5621	0.7258	1.0129	1.5054	0.6694	0.9467
				0.0911	0.0467	0.1026	0.0485	0.0241	0.0446
			CS-2	1.5802	0.7390	1.0328	1.5227	0.6781	0.9603
				0.0958	0.0507	0.1216	0.0478	0.0229	0.0446
			CS-3	1.5724	0.7354	1.0135	1.5203	0.6751	0.9496
				0.0901	0.0454	0.1006	0.0487	0.0217	0.0444
		45	CS-1	1.5640	0.7362	1.0127	1.5153	0.6815	0.9481
				0.0785	0.0455	0.0938	0.0416	0.0209	0.0404
			CS-2	1.5670	0.7353	1.0163	1.5243	0.6851	0.9642
				0.0800	0.0371	0.0828	0.0454	0.0181	0.0400
			CS-3	1.5703	0.7227	0.9967	1.5304	0.6761	0.9499
				0.0764	0.0374	0.0744	0.0435	0.0187	0.0380
	0.10	40	CS-1	1.5721	0.7215	1.0028	1.5170	0.6670	0.9463
				0.0912	0.0475	0.0999	0.0489	0.0242	0.0448
			CS-2	1.5791	0.7326	1.1091	1.5284	0.6761	0.9613
				0.0959	0.0547	0.1231	0.0481	0.0234	0.0449
			CS-3	1.5802	0.7414	1.0232	1.5284	0.6774	0.9625
				0.0933	0.0529	0.1117	0.0489	0.0247	0.0510
		45	CS-1	1.5911	0.7500	1.0293	1.5367	0.6903	0.9661
				0.0899	0.0476	0.1030	0.0463	0.0222	0.0461
			CS-2	1.5804	0.7370	1.0121	1.5364	0.6836	0.9589
				0.0820	0.0425	0.0854	0.0460	0.0209	0.0421
			CS-3	1.5574	0.7212	0.9918	1.5218	0.6739	0.9514
				0.0784	0.0377	0.0778	0.0436	0.0192	0.0459
(80,70)	0.05	60	CS-1	1.5624	0.7388	0.9833	1.5228	0.6963	0.9380
				0.0621	0.0332	0.0634	0.0400	0.0196	0.0344
			CS-2	1.5517	0.7284	0.9719	1.5169	0.6923	0.9331
				0.0547	0.0246	0.0475	0.0361	0.0157	0.0304
			CS-3	1.5458	0.7330	0.9786	1.5116	0.6926	0.9387
				0.0517	0.0288	0.0526	0.0346	0.0168	0.0315
		65	CS-1	1.5520	0.7311	0.9772	1.5187	0.6960	0.9396
				0.0552	0.0278	0.0530	0.0362	0.0176	0.0323
			CS-2	1.5397	0.7179	0.9584	1.5134	0.6890	0.9291
				0.0518	0.0202	0.0386	0.0372	0.0136	0.0261
			CS-3	1.5506	0.7269	0.9628	1.5228	0.6955	0.9312
				0.0473	0.0208	0.0363	0.0331	0.0134	0.0232
	0.10	60	CS-1	1.5558	0.7308	0.9785	1.5193	0.6915	0.9372
				0.0591	0.0307	0.0604	0.0369	0.0182	0.0337
			CS-2	1.5449	0.7199	0.9568	1.5137	0.6828	0.9205
				0.0548	0.0261	0.0443	0.0360	0.0168	0.0250
			CS-3	1.5428	0.7221	0.9726	1.5089	0.6829	0.9329
				0.0561	0.0257	0.0515	0.0376	0.0164	0.0296
		65	CS-1	1.5442	0.7315	0.9780	1.5108	0.6963	0.9392
				0.0576	0.0283	0.0548	0.0362	0.0177	0.0323
			CS-2	1.5444	0.7208	0.9597	1.5176	0.6897	0.9290
				0.0544	0.0228	0.0412	0.0360	0.0147	0.0248
			CS-3	1.5456	0.7229	0.9628	1.5180	0.6890	0.9305
				0.0503	0.0224	0.0430	0.0346	0.0143	0.0271

6.2. Illustrative example

In this section, a real data set as discussed in Meintanis [23] is considered in support of proposed methods. Suppose Y_1 is time (in minutes) of the first kick goal scored by a team and Y_2 is the time of first goal of any home team. The data is listed in Table 9 after dividing

Table 5. AIWs and CPs of parameters α, λ_1 and λ_2 without order restriction.

$T = 0.5$				ACI			Symmetric interval			HPD interval		
(n,m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	0.9893	0.9107	1.1152	0.9704	0.8909	1.1099	0.9110	0.8092	1.0134
				0.9526	0.9583	0.9687	0.9822	0.9927	0.9918	0.9743	0.9864	0.9890
			CS-2	0.9925	0.8573	1.0492	0.9815	0.8423	1.0362	0.9193	0.7695	0.9498
				0.9506	0.9553	0.9630	0.9880	0.9934	0.9941	0.9807	0.9853	0.9915
			CS-3	0.9646	0.8575	1.0664	0.9504	0.8414	1.0389	0.8919	0.7698	0.9548
				0.8588	0.9694	0.9791	0.9842	0.9959	0.9914	0.9741	0.9860	0.9883
		45	CS-1	0.9873	0.8493	1.0417	0.9641	0.8241	1.0249	0.9014	0.7501	0.9351
				0.9432	0.9415	0.9533	0.9834	0.9916	0.9848	0.9757	0.9781	0.9842
			CS-2	0.9732	0.7951	0.9844	0.9519	0.7796	0.9610	0.8917	0.7159	0.8856
				0.9611	0.9606	0.9721	0.9807	0.9872	0.9894	0.9718	0.9813	0.9842
			CS-3	0.9310	0.7916	0.9772	0.9181	0.7750	0.9523	0.8622	0.7123	0.8796
				0.9564	0.9610	0.9717	0.9877	0.9923	0.9905	0.9725	0.9859	0.9898
	0.10	40	CS-1	0.9932	0.9183	1.1373	0.9859	0.8957	1.1131	0.9243	0.8130	1.0143
				0.9675	0.9616	0.9662	0.9882	0.9937	0.9914	0.9856	0.9819	0.9898
			CS-2	0.9992	0.8778	1.0868	0.9860	0.8536	1.0533	0.9198	0.7802	0.9674
				0.9457	0.9604	0.9681	0.9860	0.9936	0.9909	0.9722	0.9814	0.9882
			CS-3	0.9884	0.8913	1.1038	0.9668	0.8702	1.0746	0.9059	0.7929	0.9825
				0.9399	0.9587	0.9708	0.9883	0.9913	0.9978	0.9744	0.9807	0.9935
		45	CS-1	0.9851	0.8588	1.1090	0.9651	0.8276	1.0257	0.9017	0.7550	0.9403
				0.9547	0.9514	0.9588	0.9875	0.9919	0.9902	0.9788	0.9792	0.9870
			CS-2	0.9716	0.7932	0.9822	0.9580	0.7797	0.9620	0.8818	0.7168	0.8790
				0.9619	0.9663	0.9794	0.9904	0.9952	0.9957	0.9828	0.9866	0.9935
			CS-3	0.9314	0.7984	0.9836	0.9190	0.7761	0.9539	0.8686	0.7149	0.8822
				0.8294	0.9592	0.9668	0.9732	0.9916	0.9928	0.9605	0.9843	0.9867
(80,70)	0.05	60	CS-1	0.8604	0.7529	0.9141	0.8483	0.7323	0.8950	0.7967	0.6727	0.8240
				0.9460	0.9693	0.9633	0.9484	0.9918	0.9855	0.9407	0.9800	0.9836
			CS-2	0.8632	0.7296	0.8774	0.8463	0.6981	0.8447	0.7940	0.6435	0.7800
				0.9365	0.9669	0.9608	0.9552	0.9914	0.9868	0.9463	0.9878	0.9865
			CS-3	0.8598	0.7490	0.8890	0.8382	0.7130	0.8624	0.7861	0.6559	0.7953
				0.9454	0.9743	0.9659	0.9672	0.9984	0.9926	0.9537	0.9917	0.9906
		65	CS-1	0.8558	0.7251	0.8735	0.8307	0.6926	0.8471	0.7794	0.6361	0.7787
				0.9293	0.9544	0.9442	0.9442	0.9895	0.9745	0.9276	0.9763	0.9669
			CS-2	0.8436	0.6786	0.8136	0.8203	0.6580	0.7937	0.7693	0.6074	0.7346
				0.9194	0.9663	0.9656	0.9453	0.9894	0.9936	0.9411	0.9781	0.9870
			CS-3	0.8416	0.7271	0.8360	0.8219	0.6730	0.8147	0.7692	0.6202	0.7519
				0.9335	0.9754	0.9639	0.9532	0.9940	0.9907	0.9443	0.9885	0.9869
	0.10	60	CS-1	0.8761	0.7699	0.9267	0.8506	0.7436	0.9006	0.7969	0.6811	0.8259
				0.9125	0.9566	0.9525	0.9463	0.9934	0.9926	0.9328	0.9854	0.9850
			CS-2	0.8719	0.6406	0.8950	0.8539	0.6986	0.8471	0.7958	0.6435	0.7801
				0.9356	0.9591	0.9605	0.9543	0.9943	0.9920	0.9366	0.9845	0.9886
			CS-3	0.8658	0.7585	0.8933	0.8406	0.7174	0.8644	0.7895	0.6561	0.7977
				0.9304	0.9576	0.9555	0.9415	0.9894	0.9864	0.9299	0.9738	0.9756
		65	CS-1	0.8636	0.7316	0.8840	0.8313	0.7000	0.8482	0.7796	0.6427	0.7796
				0.9504	0.9651	0.9557	0.9463	0.9855	0.9790	0.9354	0.9773	0.9728
			CS-2	0.8599	0.6808	0.8346	0.8323	0.6616	0.7946	0.7697	0.6132	0.7362
				0.9234	0.9775	0.9736	0.9394	0.9916	0.9906	0.9313	0.9903	0.9860
			CS-3	0.8595	0.7314	0.8460	0.8222	0.6763	0.8153	0.7692	0.6228	0.7505
				0.9141	0.9718	0.9665	0.9587	0.9936	0.9921	0.9495	0.9914	0.9889

it by 90. Here we take Y_1 and Y_2 as two competing risks under cause 1 and cause 2, respectively. In Table 9, we also present the associated competing risks data. Note that for equal failure times in data, the cause of failure is leveled as 1 to keep the number of failures due to causes 1 and 2 balanced. Wang [40] also studied these data sets. From his work Kolmogorov-Simirnov (KS) distance and the corresponding p-values for cause 1 are 0.0823

Table 6. AIWs and CPs of parameters α, λ_1 and λ_2 with order restriction.

$T = 0.5$				ACI			Symmetric interval			HPD interval		
(n,m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	0.9192	0.8775	1.0875	0.8933	0.6876	0.8621	0.7958	0.6354	0.7346
				0.9345	0.9591	0.9529	0.9466	0.9247	0.9329	0.9262	0.9170	0.9146
			CS-2	0.9292	0.8333	1.0278	0.8917	0.6560	0.8330	0.7990	0.6076	0.7055
				0.9351	0.9593	0.9675	0.9530	0.9299	0.9457	0.9372	0.9245	0.9186
			CS-3	0.9145	0.8318	1.0218	0.8897	0.6612	0.8322	0.7973	0.6119	0.7091
				0.9586	0.9698	0.9781	0.9512	0.9307	0.9480	0.9275	0.9215	0.9278
		45	CS-1	0.8973	0.7893	0.9716	0.8649	0.6309	0.7816	0.7709	0.5836	0.6573
				0.9334	0.9412	0.9536	0.9483	0.9216	0.9172	0.9197	0.9137	0.9034
			CS-2	0.8933	0.7551	0.9244	0.8707	0.6133	0.7625	0.7807	0.5683	0.6407
				0.9411	0.9603	0.9726	0.9564	0.9533	0.9477	0.9318	0.9404	0.9231
			CS-3	0.8810	0.7510	0.9166	0.8543	0.6101	0.7658	0.7655	0.5654	0.6446
				0.9256	0.9615	0.9714	0.9593	0.9324	0.9458	0.9299	0.9234	0.9255
	0.10	40	CS-1	0.9228	0.8772	1.0885	0.9001	0.6880	0.8645	0.8054	0.6360	0.7359
				0.9585	0.9627	0.9628	0.9503	0.9252	0.9410	0.9324	0.9157	0.9163
			CS-2	0.9255	0.8343	1.0283	0.9028	0.6632	0.8417	0.8120	0.6160	0.7182
				0.9462	0.9628	0.9643	0.9563	0.9310	0.9493	0.9478	0.9308	0.9232
			CS-3	0.9090	0.8543	1.0497	0.8899	0.6696	0.8544	0.7982	0.6213	0.7280
				0.9359	0.9593	0.9680	0.9504	0.9331	0.9482	0.9331	0.9256	0.9259
		45	CS-1	0.8950	0.7974	0.9820	0.8652	0.6390	0.7992	0.7720	0.5914	0.6755
				0.9545	0.9512	0.9589	0.9450	0.9194	0.9277	0.9295	0.9175	0.9099
			CS-2	0.9069	0.7535	0.9128	0.8723	0.6171	0.7760	0.7881	0.5743	0.6612
				0.9519	0.9664	0.9793	0.9744	0.9498	0.9522	0.9547	0.9396	0.9424
			CS-3	0.8814	0.7560	0.9217	0.8549	0.6170	0.7775	0.7681	0.5729	0.6586
				0.9294	0.9606	0.9667	0.9398	0.9233	0.9453	0.9052	0.9166	0.9248
(80,70)	0.05	60	CS-1	0.7904	0.6820	0.8234	0.7637	0.5609	0.6741	0.6701	0.5136	0.5485
				0.9463	0.9694	0.9630	0.9372	0.9367	0.9120	0.9059	0.9335	0.8853
			CS-2	0.7932	0.6490	0.7769	0.7634	0.5447	0.6577	0.6770	0.5003	0.5396
				0.9165	0.9677	0.9596	0.9428	0.9311	0.9217	0.9261	0.9244	0.8989
			CS-3	0.7899	0.6593	0.7895	0.7545	0.5511	0.6650	0.6660	0.5053	0.5448
				0.9455	0.9752	0.9677	0.9467	0.9480	0.9217	0.9252	0.9427	0.8999
		65	CS-1	0.7858	0.6351	0.7635	0.7429	0.5302	0.6356	0.6511	0.4842	0.5109
				0.9293	0.9543	0.9447	0.9242	0.9195	0.8857	0.8929	0.9118	0.8585
			CS-2	0.7936	0.6087	0.7238	0.7469	0.5202	0.6215	0.6644	0.4789	0.5072
				0.9196	0.9669	0.9652	0.9393	0.9440	0.9131	0.9173	0.9368	0.8944
			CS-3	0.7716	0.6171	0.7360	0.7353	0.5236	0.6278	0.6516	0.4808	0.5094
				0.9332	0.9756	0.9635	0.9384	0.9558	0.9277	0.9208	0.9393	0.9038
	0.10	60	CS-1	0.7958	0.6850	0.8222	0.7649	0.5725	0.6978	0.6748	0.5251	0.5772
				0.9114	0.9588	0.9489	0.9305	0.9176	0.8951	0.9124	0.9160	0.8739
			CS-2	0.7948	0.6482	0.7733	0.7689	0.5497	0.6674	0.6871	0.5077	0.5545
				0.9366	0.9593	0.9605	0.9446	0.9324	0.9184	0.9097	0.9209	0.8882
			CS-3	0.7901	0.6569	0.7824	0.7577	0.5519	0.6780	0.6743	0.5081	0.5630
				0.9305	0.9597	0.9560	0.9321	0.9306	0.9162	0.9056	0.9248	0.8896
		65	CS-1	0.7886	0.6417	0.7641	0.7470	0.5441	0.6569	0.6610	0.5002	0.5414
				0.9502	0.9656	0.9557	0.9385	0.9305	0.9021	0.9154	0.9195	0.8826
			CS-2	0.7996	0.6208	0.7346	0.7554	0.5311	0.6412	0.6748	0.4902	0.5298
				0.9235	0.9772	0.9736	0.9471	0.9526	0.9226	0.9260	0.9494	0.9028
			CS-3	0.7796	0.6218	0.7365	0.7396	0.5267	0.6432	0.6587	0.4859	0.5303
				0.9147	0.9725	0.9671	0.9553	0.9418	0.9236	0.9295	0.9367	0.8977

and 0.9558 and for cause 2 as 0.1018 and 0.8176, respectively. The author also presented graphical illustrations and suggested that considered data sets fit Kumaraswamy distribution good. Further, it is of important to verify the equivalence of shape parameters α_1 and α_2. So, we consider the procedure of hypothesis testing, as discussed in Section 5, to test

Table 7. AIWs and CPs of parameters α, λ_1 and λ_2 without order restriction.

$T = 0.6$				ACI			Symmetric interval			HPD interval		
(n,m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	0.9854	0.8764	1.0919	0.9585	0.8567	1.0721	0.9012	0.7807	0.9823
				0.9332	0.9443	0.9685	0.9803	0.9908	0.9880	0.9673	0.9763	0.9861
			CS-2	0.9958	0.8621	1.0594	0.9713	0.8444	1.0479	0.9129	0.7728	0.9627
				0.9296	0.9637	0.9585	0.9818	0.9882	0.9849	0.9715	0.9806	0.9843
			CS-3	0.9844	0.8779	1.0676	0.9552	0.8459	1.0451	0.8970	0.7738	0.9587
				0.9308	0.9625	0.9524	0.9753	0.9937	0.9911	0.9704	0.9825	0.9807
		45	CS-1	0.9851	0.8496	1.0306	0.9545	0.8195	1.0163	0.8936	0.7478	0.9300
				0.9395	0.9554	0.9618	0.9813	0.9943	0.9862	0.9762	0.9829	0.9872
			CS-2	0.9721	0.8095	0.9897	0.9439	0.7846	0.9679	0.8854	0.7207	0.8916
				0.9175	0.9706	0.9674	0.9825	0.9943	0.9937	0.9745	0.9856	0.9896
			CS-3	0.9569	0.7907	0.9852	0.9287	0.7728	0.9519	0.8723	0.7107	0.8783
				0.9343	0.9645	0.9663	0.9856	0.9912	0.9929	0.9734	0.9816	0.9933
	0.10	40	CS-1	0.9911	0.8993	1.0934	0.9751	0.8724	1.0849	0.9139	0.7928	0.9881
				0.9385	0.9506	0.9592	0.9793	0.9952	0.9931	0.9737	0.9815	0.9862
			CS-2	0.9952	0.8964	1.0866	0.9857	0.8581	1.0604	0.9237	0.7835	0.9725
				0.9477	0.9679	0.9768	0.9852	0.9902	0.9901	0.9751	0.9874	0.9912
			CS-3	0.9881	0.8785	1.0771	0.9563	0.8598	1.0569	0.8976	0.7857	0.9700
				0.9262	0.9507	0.9666	0.9801	0.9903	0.9932	0.9705	0.9737	0.9875
		45	CS-1	0.9874	0.8634	1.0554	0.9548	0.8394	1.0319	0.8938	0.7679	0.9468
				0.9176	0.9538	0.9633	0.9840	0.9932	0.9905	0.9733	0.9868	0.9885
			CS-2	0.9789	0.8138	0.9963	0.9470	0.7907	0.9680	0.8893	0.7278	0.8918
				0.9249	0.9583	0.9594	0.9855	0.9901	0.9849	0.9806	0.9796	0.9817
			CS-3	0.9615	0.7996	0.9898	0.9291	0.7847	0.9581	0.8725	0.7217	0.8832
				0.9591	0.9604	0.9637	0.9888	0.9956	0.9870	0.9804	0.9873	0.9821
(80,70)	0.05	60	CS-1	0.8562	0.7493	0.8995	0.8385	0.7280	0.8838	0.7889	0.6713	0.8162
				0.9255	0.9636	0.9543	0.9472	0.9902	0.9842	0.9396	0.9812	0.9783
			CS-2	0.8542	0.7216	0.8589	0.8352	0.6932	0.8388	0.7848	0.6407	0.7775
				0.9358	0.9768	0.9632	0.9562	0.9941	0.9863	0.9459	0.9891	0.9807
			CS-3	0.8565	0.7482	0.8976	0.8372	0.7130	0.8641	0.7836	0.6543	0.7942
				0.9318	0.9610	0.9543	0.9521	0.9887	0.9847	0.9391	0.9788	0.9759
		65	CS-1	0.8556	0.7461	0.8833	0.8231	0.6895	0.8395	0.7714	0.6346	0.7737
				0.9239	0.9731	0.9653	0.9416	0.9915	0.9833	0.9236	0.9866	0.9809
			CS-2	0.8534	0.7064	0.8420	0.8271	0.6600	0.7976	0.7747	0.6088	0.7374
				0.9146	0.9805	0.9692	0.9434	0.9958	0.9875	0.9406	0.9921	0.9827
			CS-3	0.8328	0.7146	0.8387	0.8238	0.6701	0.8068	0.7716	0.6175	0.7448
				0.9481	0.9768	0.9758	0.9544	0.9953	0.9968	0.9473	0.9879	0.9924
	0.10	60	CS-1	0.8738	0.7830	0.9203	0.8392	0.7308	0.8870	0.7890	0.6718	0.8181
				0.9214	0.9683	0.9596	0.9522	0.9905	0.9896	0.9392	0.9849	0.9792
			CS-2	0.8914	0.7414	0.8831	0.8416	0.7001	0.8420	0.7909	0.6459	0.7781
				0.9296	0.9605	0.9599	0.9577	0.9920	0.9883	0.9498	0.9799	0.9854
			CS-3	0.8755	0.7579	0.8982	0.8375	0.7133	0.8642	0.7847	0.6549	0.7943
				0.9119	0.9625	0.9661	0.9355	0.9922	0.9926	0.9263	0.9854	0.9826
		65	CS-1	0.8823	0.7543	0.9006	0.8410	0.7148	0.8713	0.7858	0.6532	0.7956
				0.9373	0.9642	0.9463	0.9364	0.9939	0.9805	0.9203	0.9804	0.9775
			CS-2	0.8650	0.6967	0.8403	0.8273	0.6679	0.8015	0.7755	0.6172	0.7421
				0.9258	0.9681	0.9596	0.9448	0.9916	0.9818	0.9376	0.9864	0.9808
			CS-3	0.8508	0.7202	0.8452	0.8239	0.6707	0.8069	0.7631	0.6200	0.7453
				0.9391	0.9754	0.9636	0.9464	0.9921	0.9883	0.9356	0.9881	0.9847

the equivalence with 0.05 level of significance. Likelihood ratio statistic and corresponding p-value are obtained as 2.09606 and 0.14768, respectively. These results suggest that the null hypothesis, i.e. $H_0 : \alpha_1 = \alpha_2$ may not be rejected. Therefore, for the considered football data set, the shape parameters α_1 and α_2 may be treated as equal.

Table 8. AIWs and CPs of parameters α, λ_1 and λ_2 with order restriction.

$T = 0.6$				ACI			Symmetric interval			HPD interval		
(n,m)	u	k	CS	α	λ_1	λ_2	α	λ_1	λ_2	α	λ_1	λ_2
(60,50)	0.05	40	CS-1	0.9098	0.8364	1.0437	0.8791	0.6597	0.8249	0.7794	0.6088	0.6926
				0.9334	0.9423	0.9601	0.9223	0.8997	0.9051	0.8967	0.8933	0.8896
			CS-2	0.9363	0.8368	1.0372	0.8950	0.6577	0.8210	0.8029	0.6089	0.6936
				0.9296	0.9651	0.9535	0.9440	0.9239	0.9364	0.9282	0.9168	0.9105
			CS-3	0.9347	0.8322	1.0227	0.8811	0.6571	0.8303	0.7914	0.6083	0.7085
				0.9291	0.9636	0.9524	0.9437	0.9292	0.9394	0.9175	0.9163	0.9154
		45	CS-1	0.9052	0.7902	0.9712	0.8649	0.6306	0.7806	0.7713	0.5828	0.6572
				0.9396	0.9551	0.9613	0.9532	0.9326	0.9398	0.9382	0.9266	0.9174
			CS-2	0.9222	0.7621	0.9303	0.8687	0.6174	0.7729	0.7801	0.5727	0.6515
				0.9171	0.9704	0.9676	0.9478	0.9509	0.9516	0.9281	0.9434	0.9321
			CS-3	0.8969	0.7507	0.9153	0.8586	0.6100	0.7614	0.7717	0.5653	0.6423
				0.9346	0.9640	0.9673	0.9557	0.9467	0.9451	0.9326	0.9385	0.9218
	0.10	40	CS-1	0.9100	0.8396	1.0413	0.8881	0.6659	0.8468	0.7917	0.6156	0.7193
				0.9395	0.9503	0.9476	0.9348	0.9172	0.9314	0.9244	0.9075	0.9102
			CS-2	0.9448	0.8346	1.0268	0.9074	0.6678	0.8473	0.8163	0.6194	0.7224
				0.9477	0.9674	0.9733	0.9453	0.9414	0.9524	0.9308	0.9332	0.9307
			CS-3	0.9381	0.8497	1.0421	0.8839	0.6652	0.8558	0.7955	0.6166	0.7317
				0.9289	0.9495	0.9659	0.9456	0.9151	0.9429	0.9273	0.9102	0.9197
		45	CS-1	0.9071	0.8166	0.9995	0.8762	0.6515	0.8184	0.7845	0.6033	0.6965
				0.9162	0.9533	0.9596	0.9510	0.9399	0.9453	0.9292	0.9245	0.9218
			CS-2	0.9291	0.7733	0.9363	0.8844	0.6300	0.7887	0.7988	0.5847	0.6727
				0.9269	0.9338	0.9358	0.9595	0.9378	0.9472	0.9345	0.9266	0.9234
			CS-3	0.9016	0.7571	0.9190	0.8647	0.6220	0.7798	0.7792	0.5771	0.6626
				0.9591	0.9618	0.9637	0.9598	0.9393	0.9468	0.9432	0.9303	0.9132
(80,70)	0.05	60	CS-1	0.8060	0.6796	0.8170	0.7651	0.5666	0.6825	0.6718	0.5190	0.5589
				0.9256	0.9644	0.9527	0.9258	0.9267	0.9025	0.9019	0.9168	0.8823
			CS-2	0.8042	0.6510	0.7784	0.7656	0.5485	0.6628	0.6799	0.5041	0.5464
				0.9358	0.9763	0.9637	0.9396	0.9407	0.9197	0.9171	0.9338	0.8915
			CS-3	0.7865	0.6577	0.7872	0.7494	0.5498	0.6637	0.6635	0.5043	0.5442
				0.9311	0.9626	0.9547	0.9382	0.9186	0.8997	0.9155	0.9131	0.8799
		65	CS-1	0.7855	0.6358	0.7631	0.7411	0.5327	0.6383	0.6518	0.4868	0.5159
				0.9233	0.9737	0.9657	0.9372	0.9395	0.9153	0.8971	0.9278	0.8899
			CS-2	0.7834	0.6064	0.7220	0.7489	0.5201	0.6240	0.6641	0.4776	0.5085
				0.9146	0.9804	0.9698	0.9355	0.9607	0.9233	0.9093	0.9518	0.9007
			CS-3	0.7828	0.6146	0.7287	0.7449	0.5257	0.6282	0.6617	0.4833	0.5161
				0.9481	0.9765	0.9755	0.9514	0.9568	0.9327	0.9318	0.9459	0.9208
	0.10	60	CS-1	0.8139	0.6808	0.8187	0.7659	0.5672	0.6921	0.6722	0.5199	0.5695
				0.9224	0.9696	0.9599	0.9372	0.9246	0.9131	0.9177	0.9120	0.8869
			CS-2	0.8114	0.6496	0.7718	0.7725	0.5491	0.6709	0.6888	0.5060	0.5588
				0.9316	0.9615	0.9595	0.9482	0.9253	0.9126	0.9277	0.9214	0.8962
			CS-3	0.7956	0.6564	0.7871	0.7497	0.5512	0.6706	0.6643	0.5079	0.5522
				0.9105	0.9637	0.9670	0.9241	0.9256	0.9162	0.8883	0.9218	0.8916
		65	CS-1	0.7923	0.6436	0.7700	0.7497	0.5431	0.6521	0.6530	0.4980	0.5328
				0.9271	0.9646	0.9466	0.9273	0.9275	0.9087	0.8957	0.9223	0.8757
			CS-2	0.7951	0.6167	0.7305	0.7535	0.5266	0.6368	0.6747	0.4867	0.5266
				0.9264	0.9685	0.9592	0.9463	0.9376	0.9206	0.9153	0.9357	0.8988
			CS-3	0.7908	0.6196	0.7348	0.7452	0.5269	0.6386	0.6619	0.4866	0.5271
				0.9394	0.9765	0.9631	0.9367	0.9578	0.9281	0.9115	0.9452	0.8997

Under GPHC, three partially observed data are generated and tabulated in Table 10. Due to unavailability of partially observed competing risks data, failure causes of some of the randomly chosen data are assigned as missing indicated with '$\star$'. For fixed value $(n, m) = (37, 30)$, the parameters k and T are chosen so that the three cases of GPHC data are properly described. Also three different censoring schemes are considered similar to

Table 9. Complete competing risks failure data (Y_1, Y_2) for the proportion time of football game.

Original football game proportion time					
[0.0222, 0.0222]	[0.4556, 0.0333]	[0.4667, 0.0333]	[0.6000, 0.0778]	[0.0889, 0.0889]	[0.2778, 0.1000]
[0.6111, 0.1222]	[0.4889, 0.1444]	[0.2444, 0.1556]	[0.2778, 0.1556]	[0.7111, 0.1667]	[0.1778, 0.1778]
[0.1778, 0.8333]	[0.7000, 0.2000]	[0.2000, 0.2000]	[0.2111, 0.2111]	[0.2889, 0.2222]	[0.2667, 0.2667]
[0.2889, 0.5333]	[0.3000, 0.5222]	[0.3111, 0.3111]	[0.5667, 0.3111]	[0.4889, 0.3333]	[0.3778, 0.3778]
[0.4000, 0.5778]	[0.4333, 0.4333]	[0.5889, 0.4333]	[0.4444, 0.4444]	[0.4667, 0.4667]	[0.9111, 0.5333]
[0.5444, 0.5444]	[0.5444, 0.5444]	[0.7333, 0.6889]	[0.8444, 0.7111]	[0.7333, 0.9444]	[0.7667, 0.7889]
[0.8000, 0.8000]					
Complete competing risks data					
[0.0222, 1]	[0.0333, 2]	[0.0333, 2]	[0.0778, 2]	[0.0889, 1]	[0.1000, 2]
[0.1222, 2]	[0.1444, 2]	[0.1556, 2]	[0.1556, 2]	[0.1667, 2]	[0.1778, 1]
[0.1778, 1]	[0.2000, 2]	[0.2000, 1]	[0.2111, 1]	[0.2222, 2]	[0.2667, 1]
[0.2889, 1]	[0.3000, 1]	[0.3111, 1]	[0.3111, 2]	[0.3333, 2]	[0.3778, 1]
[0.4000, 1]	[0.4333, 1]	[0.4333, 2]	[0.4444, 1]	[0.4667, 1]	[0.5333, 2]
[0.5444, 1]	[0.5444, 1]	[0.6889, 2]	[0.7111, 2]	[0.7333, 1]	[0.7667, 1]
[0.8000, 1]					

Table 10. Partially observed competing risks GPHC data for the proportion time of football game.

Data I: $n = 37, m = 30, k = 25, T = 0.3, d = 25$ and $R = (0^{*29}, 7)$					
[0.0222, 1]	[0.0333, 2]	[0.0333, $\star$]	[0.0778, 2]	[0.0889, 1]	[0.1000, 2]
[0.1222, 2]	[0.1444, 2]	[0.1556, 2]	[0.1556, $\star$]	[0.1667, 2]	[0.1778, 1]
[0.1778, 1]	[0.2000, 2]	[0.2000, $\star$]	[0.2111, 1]	[0.2222, 2]	[0.2667, 1]
[0.2889, 1]	[0.3000, 1]	[0.3111, 1]	[0.3111, 2]	[0.3333, 2]	[0.3778, 1]
[0.4000, 1]					
Data II: $n = 37, m = 30, k = 26, T = 0.6, d = 28$ and $R = (7, 0^{*29})$					
[0.0222, 1]	[0.0333, 2]	[0.0778, 2]	[0.0889, 1]	[0.1000, 2]	[0.1222, 2]
[0.1444, 2]	[0.1556, 2]	[0.1667, 2]	[0.1778, 1]	[0.1778, $\star$]	[0.2000, 2]
[0.2000, $\star$]	[0.2111, 1]	[0.2222, 2]	[0.2667, 1]	[0.2889, 1]	[0.3000, 1]
[0.3111, 2]	[0.3333, 2]	[0.3778, 1]	[0.4000, 1]	[0.4333, 1]	[0.4333, $\star$]
[0.4444, 1]	[0.4667, 1]	[0.5333, 2]	[0.5444, 1]		
Data III: $n = 37, m = 30, k = 28, T = 0.8, d = 30$ and $R = (1^{*7}, 0^{*23})$					
[0.0222, 1]	[0.0333, 2]	[0.0778, 2]	[0.0889, 1]	[0.1000, 2]	[0.1222, 2]
[0.1444, 2]	[0.1556, 2]	[0.1667, 2]	[0.1778, $\star$]	[0.1778, 1]	[0.2000, 2]
[0.2000, $\star$]	[0.2111, 1]	[0.2222, 2]	[0.2667, 1]	[0.2889, 1]	[0.3000, 1]
[0.3111, $\star$]	[0.3333, 2]	[0.3778, 1]	[0.4000, 1]	[0.4333, 1]	[0.4444, 1]
[0.4667, 1]	[0.5333, 2]	[0.5444, 1]	[0.6889, 2]	[0.7111, 2]	[0.7333, 1]

thee simulation study where censoring is applied with random removal. In the first censoring scheme, censoring is done at the mth failure which is a right censoring. In the second censoring scheme, censoring occurs at the observation of first data and is sometime known as the first step censoring. Similarly, in the third censoring scheme, censoring occurs at observation of first $(n - m)$ observations with single removal at each observation. At first, point and interval estimates are obtained for complete competing risks data tabulated in Table 11. Further, all the point and interval estimates are obtained for unrestricted case under the three data sets. Estimation results are presented in the same Table 11. Various tabulated results in Table 11 suggest that for obtained parameter values the model fits data sets well. The corresponding Bayes estimates are computed against non-informative priors. As an illustration we further mention that when we compare the proposed estimate with corresponding complete data results we see that estimate obtained under censoring is satisfactorily close to the complete data estimate. So we may observe that proposed estimates

Table 11. Point and interval estimates of unknown parameters of Kumaraswamy distribution for real data set.

Complete data	α	λ_1	λ_2
MLE	1.2119	1.4486	1.2313
Bayes	0.9658	0.9647	0.9152
ACI	(0.9113, 1.5126) [0.6012]	(0.6877, 2.2095) [1.5217]	(0.5459, 1.9166) [1.3706]
BCI	(0.6407, 1.3809) [0.7401]	(0.5104, 1.6576) [1.1472]	(0.4836, 1.5590) [1.0754]
HPD	(0.6211, 1.3112) [0.6901]	(0.5005, 1.5468) [1.0462]	(0.4495, 1.4438) [0.9942]
Data I	α	λ_1	λ_2
MLE	1.2972	1.5867	1.5867
Bayes	0.8847	0.7380	0.8633
ACI	(0.8910, 1.7033) [0.8123]	(0.3527, 2.8207) [2.4680]	(0.3527, 2.8207) [2.4680]
BCI	(0.5483, 1.3342) [0.7859]	(0.3151, 1.4944) [1.1793]	(0.3607, 1.7455) [1.3848]
HPD	(0.5519, 1.2793) [0.7274]	(0.2581, 1.2686) [1.0105]	(0.3219, 1.5095) [1.1876]
Data II	α	λ_1	λ_2
MLE	1.4504	2.2888	2.1128
Bayes	1.0180	1.0642	1.1404
ACI	(1.0467, 1.8541) [0.8074]	(0.7046, 3.8731) [3.1685]	(0.6132, 3.6123) [2.9990]
BCI	(0.6476, 1.4451) [0.7974]	(0.4613, 2.0148) [1.5535]	(0.5267, 2.1901) [1.5534]
HPD	(0.6623, 1.4229) [0.7606]	(0.3704, 1.7267) [1.3563]	(0.5368, 1.9773) [1.4405]
Data III	α	λ_1	λ_2
MLE	1.3928	1.8984	1.762836
Bayes	1.0398	1.0270	1.1029
ACI	(1.0267, 1.7589) [0.7321]	(0.6955, 3.1014) [2.4059]	(0.6168, 2.9088) [2.2920]
BCI	(0.6778, 1.4654) [0.7875]	(0.4799, 1.8671) [1.3872]	(0.5444, 2.0332) [1.4887]
HPD	(0.6644, 1.3941) [0.7297]	(0.4149, 1.6653) [1.2504]	(0.5176, 1.8204) [1.3028]

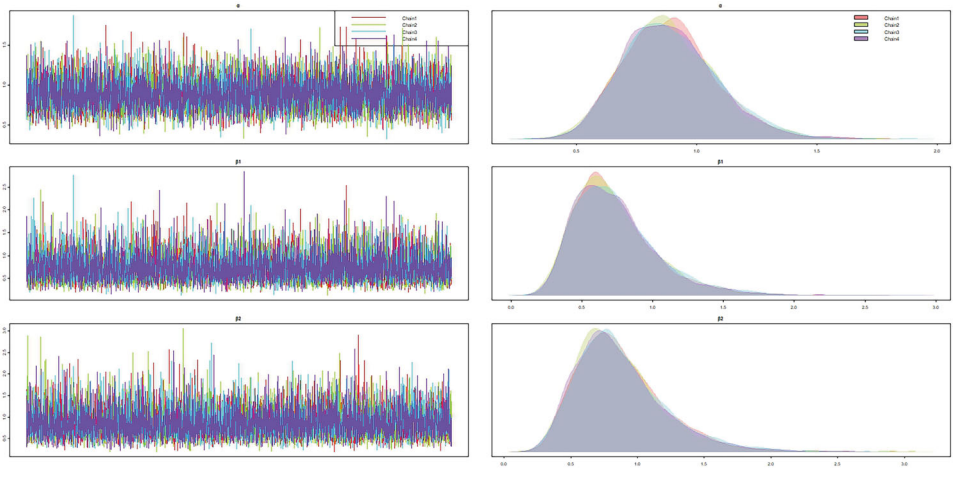


Figure 1. Trace (left) and density (right) plots of MCMC samples for the three parameters under the partially observed competing risks GPHC data I.

work well under given competing risk framework. For assessing the convergence of the MCMC algorithm for the considered data sets, we have presented trace and density plots for the MCMC samples for the three parameters (see, Figures 1–3). From these visuals we observe that plots for different chains show satisfactory behavior for all the parameters under considered data sets. Also the density plots overlap well for different chains in all

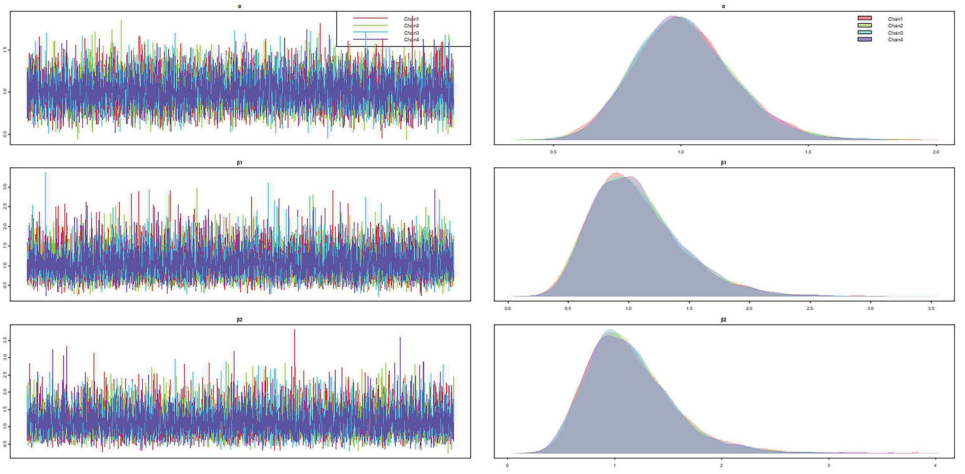


Figure 2. Trace (left) and density (right) plots of MCMC samples for the three parameters under the partially observed competing risks GPHC data II.

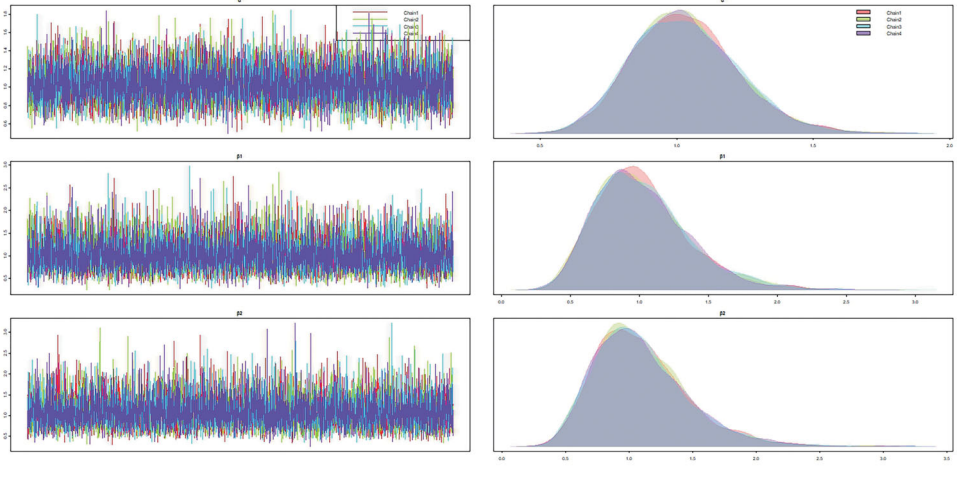


Figure 3. Trace (left) and density (right) plots of MCMC samples for the three parameters under the partially observed competing risks GPHC data III.

the plots showing that the samples come from same posterior distribution. Therefore, the trace and density plots suggest the convergence of the MCMC algorithm. The results listed in Table 11 suggest that Bayes intervals are relatively superior to corresponding asymptotic estimates in terms of interval length. We further observe that HPD intervals are better than other estimates. Also plots for profile log-likelihood function and gamma approximation of marginal posterior density of α are also presented in Figure 4. The plots suggest that profile log-likelihood is unimodal implying unique maximum and the gamma approximation of marginal posterior density of α is relatively good for considered data. In addition, as suggested by reviewers, the program code running in R software of the illustrative example is also provided in the Supplemental Material.

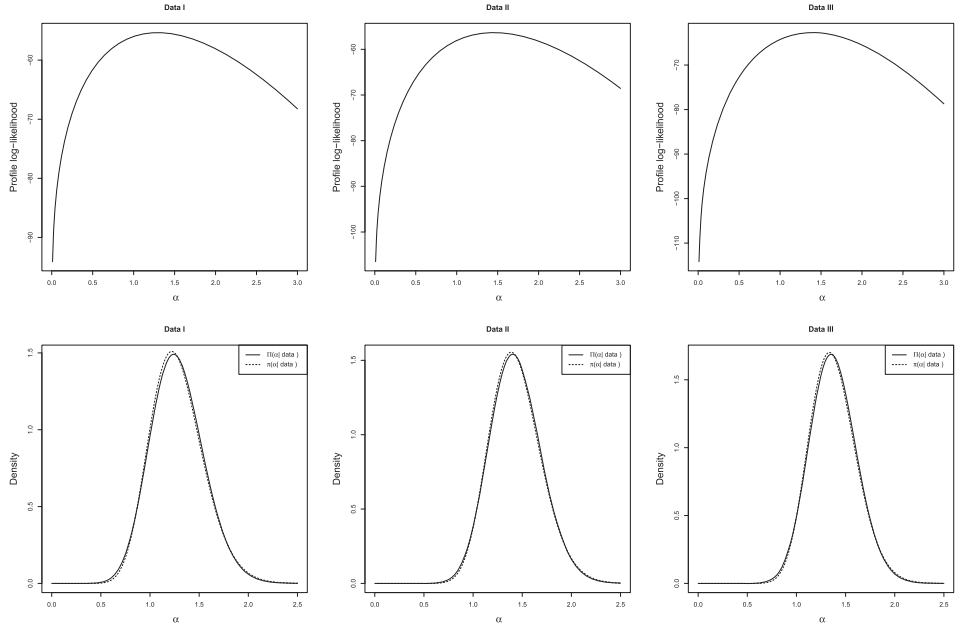


Figure 4. Plots of the profile log-likelihoods of α for real data (upper row) and gamma approximation of $\Pi(\alpha \mid \text{data})$ (lower row) for α under the three partially observed competing risks GPHC data.

7. Concluding remarks

In this paper, a generalization of competing risks problem termed as partially observed competing risks is studied when data are derived using the generalized progressive hybrid censoring. Various inferences for unknown parameters of Kumaraswamy distributions are derived under classical and Bayesian context. The point and interval estimators of parameters are evaluated against unrestricted and order restricted shape parameters of Kumaraswamy distributions. Extensive Monte Carlo simulations are performed to examine the behavior of all point and interval estimators. We also analyzed a real data set in support of studied problem. This work on GPHC can be extended to the other censoring schemes and to other probability models under partially observed competing risks data.

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